Supplementary materials for “Global hospital admissions and in-hospital mortality associated with all-cause and virus-specific acute lower respiratory infections in children and adolescents aged 5-19 years between 1995 and 2019: a systematic review and modelling study”

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Appendix 1 Search strategy

Medline (Ovid)

#	Search
	1. clinical pneumonia.mp. 2. severe pneumonia.mp. 3. alri.mp. 4. lower respiratory infection$.mp. 5. exp Pneumonia/ or pneumonia.mp. 6. exp Bronchiolitis/ or Bronchiolitis, Viral/
	7. exp Influenza, Human/ 8. exp Influenzavirus B/ or exp Influenzavirus A/ or exp Influenzavirus C/ 9. *Influenza Vaccines/ or *Influenza A virus/ or *Influenza, Human/ 10. exp Respiratory Syncytial Viruses/ or exp Respiratory Syncytial Virus Infections/ or respiratory syncytial virus.mp. 11. respiratory syncytial virus*.mp. 12. exp Parainfluenza Virus 1, Human/ or exp Parainfluenza Virus 2, Human/ or exp Parainfluenza Virus 3, Human/ or exp Parainfluenza Virus 4, Human/ or exp Parainfluenza virus infection/ 13. parainfluenza virus.mp. or piv.mp. 14. metapneumovirus.mp. or exp metapneumovirus/ or MPV.mp. 15. respiratory virus.mp.
	16. disease burden.mp. 17. exp morbidity/ or exp incidence/ or exp prevalence/ or exp proportion/ or exp general practice/ or exp outpatient/ or exp clinic /or exp hospitalization/ or exp hospital admission/ or exp hospital admission/ or exp mortality/ or exp death
18	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15
19	16 or 17
20	18 and 19
21	limit 21 to (humans and yr="1995 - 2019" and ("child (6 to 12 years)" or "adolescent (13 to 18 years)"))
22	ep.fs.
23	21 and 22

Embase (Ovid)

#	Search
	1. clinical pneumonia.mp. 2. severe pneumonia.mp. 3. alri.mp. 4. exp lower respiratory tract infection/ or lower respiratory infection$.mp. 5. exp Pneumonia/ or pneumonia.mp. 6. exp BRONCHIOLITIS/ or VIRAL BRONCHIOLITIS/
	7. exp Influenza virus A/ or exp influenza/ or exp Influenza virus/ or exp Influenza virus B/ 8. exp parainfluenza virus infection/ or para influenza virus.mp. or parainfluenza virus.mp. or parainfluenzavirus.mp. or virus,parainfluenza.mp. or PIV.mp. 9. exp metapneumovirus/ or exp metapneumovirus infection/ or mpv.mp. 10. respiratory syncytial virus.mp. or exp Respiratory syncytial pneumovirus/ or exp respiratory syncytial virus infection/ or respiratory syncytial virus*.mp. 11. respiratory virus.mp.
	12. exp incidence/ or exp prevalence/ or exp morbidity/ or exp general practice/ or exp outpatient/ or exp proportion/ or exp hospital admission/ or exp hospitalization/ or exp death/ or exp mortality/ 13. disease burden.mp.
14	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11
15	12 or 13
16	14 and 15
17	limit 16 to (human and yr="1995 - 2019" and (school child <7 to 12 years> or adolescent <13 to 17 years>))
18	ep.fs.
19	17 and 18

Global Health (Ovid)

#	Search
	1. clinical pneumonia.mp. 2. severe pneumonia.mp. 3. alri.mp. 4. exp lower respiratory tract infection/ or lower respiratory infection$.mp. 5. exp Pneumonia/ or pneumonia.mp. 6. exp BRONCHIOLITIS/ or VIRAL BRONCHIOLITIS/
	7. exp Influenza virus A/ or exp influenza/ or exp Influenza virus/ or exp Influenza virus B/ 8. exp parainfluenza virus infection/ or para influenza virus.mp. or parainfluenza virus.mp. or parainfluenzavirus.mp. or virus,parainfluenza.mp. or PIV.mp. 9. exp metapneumovirus/ or exp metapneumovirus infection/ or mpv.mp. 10. respiratory syncytial virus.mp. or exp Respiratory syncytial pneumovirus/ or exp respiratory syncytial virus infection/ or respiratory syncytial virus*.mp. 11. respiratory virus.mp.
	12. exp incidence/ or exp morbidity/ or exp hospital admission/ or exp general practice/ or exp outpatient/ or exp clinic/ or exp hospital admission/ or exp hospitalization/ or exp prevalence/ or exp proportion/ or exp mortality/ or exp death/ 13. disease burden.mp.
14	(Children and adolescents or adolescent).mp.
15	1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11
16	12 or 13
17	14 and 15 and 16
18	limit 17 to yr="1995 - 2019"

CINAHL

#	Search
	SU pneumonia OR SU pneumonia virus$ OR SU pneumonia, bacterial OR SU bronchiolitis OR SU lower respiratory infection OR SU respiratory syncytial virus OR SU influenza, human OR SU parainfluenza OR SU metapneumovirus
	MH disease burden OR MH incidence OR MH prevalence OR MH morbidity OR MH mortality OR MH hospitalization OR MH hospital admission OR MH outpatient OR MH clinic
	Limit to 1995 – 2019 Limit to child: 6-12 years and adolescent: 13-18 years

Web of Science

Topic =(lower respiratory infection OR pneumonia OR bronchiolitis OR influenza virus OR parainfluenza virus OR metapneumovirus OR respiratory syncytial virus OR respiratory aetiology)
AND Topic=(school aged OR adolescent OR teenager OR teen OR older children and adolescents)
AND Topic=(disease burden OR incidence OR prevalence OR morbidity OR mortality OR death OR hospital admission OR outpatient OR clinic OR general practice)
From 1995 to 2019
4031 papers

Global Index Medicus, including LILACS regional index.

#	Search
	Title, abstract, subject: pneumonia OR bronchiolitis OR lower respiratory tract infection OR influenza OR respiratory syncytial virus OR metapneumovirus OR parainfluenza virus AND Title, abstract, subject: children and adolescents OR adolescent AND Title, abstract, subject: incidence OR prevalence OR proportion OR hospital admission OR mortality OR disease burden 1995 - 2019

CNKI
Topic =respiratory tract infections OR pneumonia OR bronchiolitis OR influenza virus OR respiratory syncytial virus OR metapneumovirus OR parainfluenza virus OR respiratory aetiologies
AND
Topic = adolescent OR teen OR school-age OR children and adolescents
1995-2019

Wanfang
Topic = respiratory tract infections OR pneumonia OR bronchiolitis OR influenza virus OR respiratory syncytial virus OR metapneumovirus OR parainfluenza virus OR respiratory aetiologies
AND
Topic = adolescent OR teen OR school-age OR children and adolescents
AND
Topic = prevalence OR incidence OR hospital admission OR mortality
1995-2019

CQvip
Topic = respiratory tract infections OR pneumonia OR bronchiolitis OR influenza virus OR respiratory syncytial virus OR metapneumovirus OR parainfluenza virus OR respiratory aetiologies
AND
Topic = adolescent OR teen OR school-age OR children and adolescents OR primary school OR secondary school
AND
Topic = prevalence OR incidence OR hospital admission OR outpatient OR mortality OR burden OR epidemiology OR aetiological study
1995-2019

Google search (200 search results)
All these words: Child AND pneumonia
Any of these words: influenza OR respiratory syncytial virus OR metapneumovirus OR parainfluenza OR etiology
AND filetype:pdf
Time: ~ to 2019.

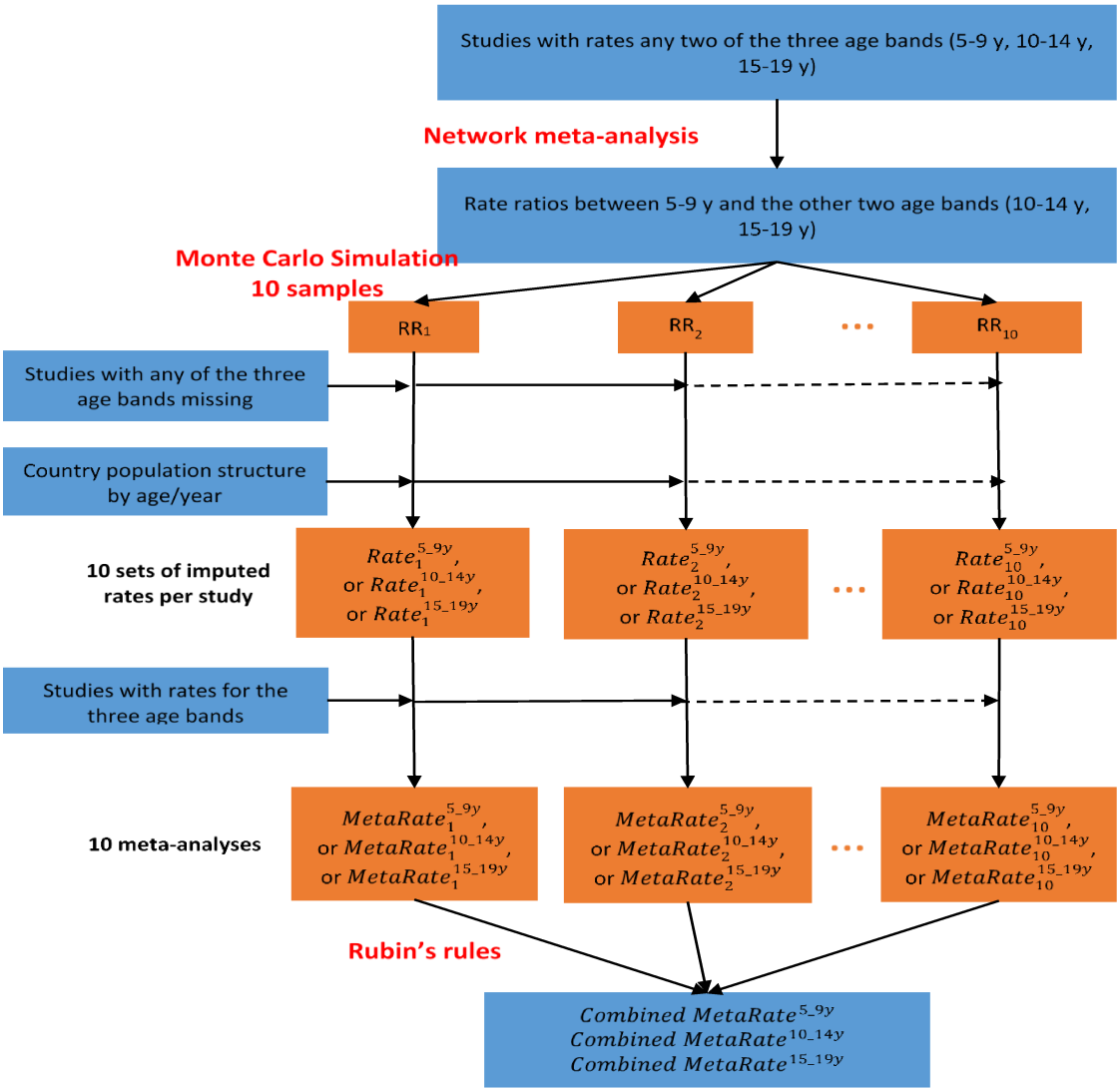
Appendix 2 Data preparation and imputation

For hospital admission rates of ALRI and influenza-associated ALRI, we extracted the number of cases and population-at-risk per study. Rates were adjusted for healthcare utilisation where available. Influenza-specific rates were additionally adjusted for levels of testing where available.

We imputed data for any of the three age bands (5-9 years, 10-14 years, or 15-19 years) where missing. Since data on rate ratios between the three age bands were only available for hospital admission rates of ALRI and influenza-associated ALRI, rates and the number of cases were imputed for the two outcomes. We used a multiple imputation approach was as used previously for influenza virus burden estimation.[1,2] The imputation was done at the study level following three steps: (1) imputing the denominator; (2) imputing the rate; (3) calculating the case number by combining the denominator and rate. Details of each step of imputation are presented below.

(1) We imputed denominator for any of the three age bands using denominators reported in each study and the country-level population structure by single year of age.[3]

(2) We imputed rates using a multiple imputation approach by assuming the rates were missing at random.[1,2] Figure S2.1 shows the imputation of rates. First, we pooled rate ratios between three age bands using the network meta-analysis, with the age band 5-9 years as the reference.[4] Second, the pooled rate ratios were assumed to follow log-normal distributions, and 10 samples of each rate ratio were simulated. Third, we estimated rate ratios for any other age bands (e.g., 5-14 years versus 5-9 years) using the rate ratios between the three age bands and the UN Population Division country population structures by single year of age (Table S3.1).[3] We assumed the hospital admission rate was the similar within each five-year age band. Fourth, with samples of rate ratios and the observed rates in each study, we calculated 10 samples of rates for each of the three age bands for each study. Fifth, numbers of cases for each study were calculated using the denominator and imputed rates. Following this strategy, 10 datasets were imputed for each of the three age bands. Sixth, meta-analysis was done for each dataset, and the meta–estimates were combined together using the Rubin's rules.[5,6]



Error! No text of specified style in document.**Figure S2.1. Imputing missing rates for any of 5-9, 10-14, and 15-19 years using the multiple imputation approach.**

Table S2.1. Estimation of rate ratios for other age bands.

Age band	Rate ratio (A)*	UNPD country population (B)[3]	Estimated rate ratio VS 5-9 y
5-9 y	1.0	B1	--
10-14 y	A2	B2	--
15-19 y	A3	B3	--
5-14 y			$\frac{(1*B1+A2*B2)}{(B1+B2)}$
5-19 y			$\frac{(1*B1+A2*B2+A3*B3)}{(B1+B2+B3)}$
Other age bands…		Corresponding B	…

Table S2.2. Comparison of rate ratios from the network meta-analysis and the conventional meta-analysis.

Age	Rate ratios from the network meta-analysis	Rate ratios from the conventional meta-analysis
ALRI hospital admissions		
5-9 y	1.0	1.0
10-14 y	0.41 (0.37-0.46)	0.44 (0.32-0.58) (9 studies)
15-19 y	0.44 (0.37-0.51)	0.52 (0.22-1.26) (4 studies)
influenza-associated ALRI hospital admissions		
5-9 y	1.0	1.0
10-14 y	0.41 (0.27-0.62)	0.41 (0.26-0.64) (8 studies)
15-19 y	0.22 (0.10-0.46)	0.25 (0.20-0.32) (2 studies)

Table S2.3 ALRI hospital admission rates and number of hospital admissions without imputation.

By World Bank income region		No imputation†	Imputation
Low	No. of countries**Error! Bookmark not defined.**	2	2
	Hospital admission rate	2.2 (1.6-3)	1.9 (1.4-2.6)
	Hospital admissions	586500 (425600-797100)	525100 (378800-718500)
Lower middle	No. of countries**Error! Bookmark not defined.**	11	11
	Hospital admission rate	3.4 (2.5-4.6)	2.9 (2.1-4.1)
	Hospital admissions	2991600 (2228900-4044400)	2608700 (1885700-3630800)
Upper middle	No. of countries**Error! Bookmark not defined.**	7	7
	Hospital admission rate	5 (3.3-7.4)	3.9 (2.6-5.6)
	Hospital admissions	2716100 (1810900-4057000)	2125400 (1456800-3092900)
High	No. of countries**Error! Bookmark not defined.**	21	21

* Estimated from the network meta-analysis.

† All data for any age bands between 5 and 19 years were analysed to generate an average rate for 5-19 years.

	Hospital admission rate	1.4 (1.3-1.6)	1.4 (1.2-1.7)
	Hospital admissions	289800 (262300-323800)	286800 (246200-340400)
Global	Hospital admissions	6584000 (4727700-9222300)‡	5546100 (3967500-7782700)

‡ The slightly higher estimate when imputation was not done could be due to the more frequent report of data for 5-9y, which were higher than older age groups (10-14y).

Appendix 3 Sensitivity analyses

1. ALRI hospital admissions

We excluded the two potential outlier studies with the highest rates (one in Bolivia and the other one in China) in a sensitivity analysis (Table S3.1). After excluding the two studies, we were unable to estimate burden in Bolivia as there were no other studies; the estimate in China after the exclusion was similar to that in the main analysis, with uncertainty ranges overlapping. The estimate of global ALRI hospital admissions remained similar after the exclusion.

Table S3.1 Estimates of ALRI hospital admission rates (per 1,000 children and adolescents per year) and hospital admissions after excluding two potential outlier studies with the highest rates.[*]

Country / region		5-9 years	10-14 years	15-19 years	5-19 years
China	No. of studies[†]	9 (9)	9 (9)	9 (9)	
	Hospital admission rate[‡]	6.2 (4.8-7.9)	2.6 (2.1-3.4)	2.7 (2.1-3.6)	
	Hospital admissions	536300 (418500-679700)	221700 (173300-280600)	225600 (172500-291600)	983500 (764300-1251800)
By World Bank income region					
Low	No. of studies	2	2	2	2
	Hospital admission rate	3 (2.2-4.1)	1.2 (0.9-1.7)	1.3 (0.9-1.8)	1.9 (1.4-2.6)
	Hospital admissions	301200 (218800-409500)	110000 (79100-150900)	99600 (70500-138700)	514000 (370800-703200)
Lower middle	No. of studies	10	10	10	10
	Hospital admission rate	4.4 (3.1-6.1)	2 (1.4-2.7)	2 (1.4-2.9)	2.8 (2-3.9)
	Hospital admissions	1331800 (960100-1863500)	586100 (423600-813900)	579800 (412100-817600)	2474200 (1778900-3462400)
Upper middle	No. of studies	7	7	7	7
	Hospital admission rate	5.4 (4.3-6.8)	2.3 (1.8-2.9)	2.3 (1.8-3)	3.4 (2.7-4.3)
	Hospital admissions	1020900 (811800-1284500)	415100 (324900-528500)	418700 (324500-538600)	1847600 (1455600-2342900)
High	No. of studies	21	21	21	21
	Hospital admission rate	2.3 (2-2.7)	1 (0.8-1.1)	1 (0.8-1.2)	1.4 (1.2-1.7)
	Hospital admissions	154500 (135400-179800)	64500 (54900-77000)	68600 (56800-83900)	286900 (246400-340000)
Global	Hospital admissions	2808400 (2126100-3737200)	1175700 (882600-1570300)	1166600 (864000-1578700)	5122800 (3851600-6848500)

[*]Estimates in countries other than China and Bolivia were not presented in this table as they were the same with those in the main analysis. We were unable to estimate burden in Bolivia as there were no data after excluding the two studies.

[†] The number in parentheses is the number of imputed studies.

[‡] Estimates of hospital admission rates were from meta-analyses.

2. ALRI in-hospital deaths

Table S3.2 Estimates of ALRI in-hospital case-fatality ratios (hCFRs, %) and in-hospital mortality by World Bank income region.[*]

	World Bank income region	No. of studies	CFR%	In-hospital mortality
Main analysis – including all CFR data[†]	Low & lower middle	7	1.6 (0.7-4.1)	51100 (18300-126600)
	Upper middle	8	1.6 (0.6-4.2)	32500 (11900-90400)
	High	12	0.4 (0.2-0.6)	1000 (600-1800)
	Global			87900 (40300-180600)
Using CFR data from a mixture of 5-14 y and 5-19 y[‡]	Low & lower middle	7	1.6 (0.7-4.1)	51100 (18300-126600)
	Upper middle	5	1.7 (0.5-5.2)	34800 (11100-109900)
	High	11	0.4 (0.2-0.7)	1000 (600-1900)
	Global			90900 (41200-197300)
Using CFR data for 5-19 y[§]	Low & lower middle	3	2.1 (0.9-5.0)	64900 (23800-157100)
	Upper middle	3	0.2 (0.1-0.6)	9700 (6600-14200)
	High	6	0.5 (0.4-0.8)	1000 (500-2200)
	Global			75100 (33700-169000)

[*] CFR estimates were from meta-analyses. In-hospital mortality were estimated by combining CFR estimates and the ALRI hospital admissions for 5-19 years.

[†] Based on all 27 studies. Of 27 studies, 24 studies reported data for 5-14 years or 5-19 years, the other studies reported data for 5-9 years (2 studies), 6-15 years (1 study), and 5-12 years (1 study).

[‡] Based on data from a mixture of 5-14 years and 5-19 years.

[§] Based on data for 5-19 years.

3. Influenza-associated ALRI hospital admissions

Table S3.3 Estimates of influenza-associated ALRI hospital admissions after excluding two potentially outlier studies with the highest rates.*

Country / region		5-9 years	10-14 years	15-19 years	5-19 years
China	No. of data points	4 (4)	4 (4)	4 (4)	NA
	Hospital admission rates (/1,000)	1 (0.7-1.3)	0.4 (0.2-0.6)	0.2 (0.1-0.5)	NA
	Hospital admissions	82700 (61500-109900)	32600 (19600-53100)	16000 (5800-42200)	131300 (86800-205200)
By World Bank income region					
Low	No. of countries	2	2	2	2
	Hospital admission rates (/1,000)	0.4 (0.3-0.5)	0.2 (0.1-0.2)	0.1 (0-0.2)	0.2 (0.1-0.3)
	Hospital admissions	38900 (27700-54000)	14100 (9000-21700)	5900 (2700-12500)	59500 (39800-88900)
Lower middle	No. of countries	8	8	8	8
	Hospital admission rates (/1,000)	0.9 (0.4-2.3)	0.4 (0.1-1.2)	0.2 (0.1-0.8)	0.5 (0.2-1.5)
	Hospital admissions	280900 (112500-715700)	112600 (36200-366200)	67500 (19100-234000)	452800 (164100-1297900)
Upper middle	No. of countries	3	3	3	3
	Hospital admission rates (/1,000)	0.9 (0.7-1.2)	0.4 (0.2-0.6)	0.2 (0.1-0.5)	0.5 (0.3-0.8)
	Hospital admissions	172700 (127100-232900)	68500 (40700-115300)	33300 (12200-92600)	273200 (178800-439400)
High	No. of countries	13	13	13	13
	Hospital admission rates (/1,000)	0.2 (0.2-0.3)	0.1 (0.1-0.2)	0 (0-0.1)	0.1 (0.1-0.2)
	Hospital admissions	16200 (11700-23300)	6900 (4300-11100)	3400 (1600-6900)	26400 (17600-41200)
Global	Hospital admissions	508800 (278900-1025900)	202000 (90000-514300)	110000 (35600-346000)	811800 (400300-1867400)

*Estimates in countries other than China and Bolivia were not presented in this table as they were the same with those in the main analysis. We were unable to estimate burden in Bolivia as there were no data after excluding the two studies.

Table S3.4 Estimates of influenza-associated ALRI hospital admission rates (per 1,000 children and adolescents per year) and hospital admissions after excluding studies with a high risk in case definition.*

Region		5-9 y	10-14 y	15-19 y	5-19 y
By World Bank income region					
Low	No. of countries	2	2	2	2
	Hospital admission rates (/1,000)	0.4 (0.3-0.5)	0.2 (0.1-0.2)	0.1 (0-0.2)	0.2 (0.1-0.3)
	Hospital admissions	38900 (27700-54000)	14100 (9000-21700)	5900 (2700-12500)	59500 (39800-88900)
Lower middle	No. of countries	9	9	9	9
	Hospital admission rates (/1,000)	1 (0.4-2.4)	0.4 (0.1-1.3)	0.2 (0.1-0.9)	0.5 (0.2-1.5)
	Hospital admissions	295700 (121000-740000)	118600 (39000-377900)	70200 (19700-246600)	475800 (175700-1346000)
Upper middle	No. of countries	3	3	3	3
	Hospital admission rates (/1,000)	1.7 (0.8-3.6)	0.7 (0.3-1.7)	0.4 (0.1-1.1)	0.9 (0.4-2.1)
	Hospital admissions	321800 (147400-679100)	129100 (53600-301400)	64500 (21400-191000)	512900 (221200-1166700)
High	No. of countries	11	11	11	11
	Hospital admission rates (/1,000)	0.3 (0.2-0.3)	0.1 (0.1-0.2)	0.1 (0-0.1)	0.1 (0.1-0.2)
	Hospital admissions	17800 (13800-22900)	7500 (5000-11200)	3800 (1900-7500)	28900 (20500-41500)
Global	Hospital admissions	674100 (309900-1496000)	269200 (106500-712100)	144300 (45600-457600)	1077100 (457200-2643100)

* High risk of bias in case definition: ICD-coded influenza-confirmed respiratory hospital admissions (e.g., ICD-10 J09-10); hospitalised with fever or acute respiratory infection AND lab-confirmed influenza.

Table S3.5 Estimates of influenza-associated ALRI hospital admission rates (per 1,000 children and adolescents per year) and hospital admissions after excluding studies with unknown testing levels

Region		5-9 y	10-14 y	15-19 y	5-19 y
By World Bank income region					
Low	No. of countries	1	1	1	1
	Hospital admission rates (/1,000)	0 (0-0)	0 (0-0)	0 (0-0)	0 (0-0)
	Hospital admissions	3800 (3400-4300)	1400 (1000-1900)	600 (300-1200)	5900 (4800-7400)
Lower middle	No. of countries	8	8	8	8
	Hospital admission rates (/1,000)	1 (0.4-2.5)	0.4 (0.1-1.3)	0.3 (0.1-0.9)	0.6 (0.2-1.6)
	Hospital admissions	305800 (124700-767600)	122100 (40000-390600)	72300 (20300-253500)	490500 (180600-1391000)
Upper middle	No. of countries	3	3	3	3
	Hospital admission rates (/1,000)	1.7 (0.8-3.6)	0.7 (0.3-1.7)	0.4 (0.1-1.1)	0.9 (0.4-2.1)
	Hospital admissions	321800 (147400-679100)	129100 (53600-301400)	64500 (21400-191000)	512900 (221200-1166700)
High	No. of countries	6	6	6	6
	Hospital admission rates (/1,000)	23900 (17600-32100)	10000 (6500-15300)	5000 (2400-10000)	38500 (26200-57000)
	Hospital admissions	0.4 (0.3-0.5)	0.1 (0.1-0.2)	0.1 (0-0.1)	0.2 (0.1-0.3)
Global	Hospital admissions	655300 (293100-1483100)	262700 (101100-709300)	142400 (44400-455800)	1047700 (432700-2622200)

Table S3.6 Estimates of proportion positives of influenza-associated ALRI and the hospital admissions in children and adolescents aged 5-19 years by World Bank income region. [*]

World Bank income region	No. of studies	Proportion[†]	Influenza-associated ALRI hospital admissions
Low & lower middle	31	19.2 (14.4-25.6)	594500 (381800-908600)
Upper middle	26	8 (5.4-11.9)	169200 (94600-287500)
High	13	10.9 (6.3-18.7)	30800 (17600-55400)
Global			796600 (525900-1177500)

[*] Based on all 70 studies. Of 70 studies, three studies for 5-9 years, 50 studies for 5-14 years, and 17 studies for 5-19 years.

[†] Estimates from meta-analyses.

3. Respiratory syncytial virus-associated ALRI hospital admissions

Table S3.7 Estimates of proportion positives of respiratory syncytial virus-associated ALRI and the hospital admissions in children and adolescents aged 5-19 years by World Bank income region.

	World Bank income region	No. of studies	Proportion*	Respiratory syncytial virus-associated ALRI hospital admissions
The main analysis – including all proportion data†	Low & lower middle	15	4.0 (2.4-6.8)	125000 (66800-222600)
	Upper middle	27	4.5 (2.8-7.2)	94500 (51100-170200)
	High	12	3.9 (1.6-9.2)	10800 (4800-27600)
	Global			231800 (142700-373200)
Using proportion data from a mixture of 5-14 y and 5-19 y‡	Low & lower middle	15	4 (2.4-6.6)	123200 (66500-218400)
	Upper middle	26	4.3 (2.7-6.9)	90700 (48900-165000)
	High	10	3.4 (1.3-9.3)	9500 (3800-28100)
	Global			227100 (138700-367300)
Using proportion data for 5-19 y§	Low & lower middle	6	4.5 (1.9-10.9)	141200 (52000-341200)
	Upper middle	7	7.8 (3.2-18.7)	161800 (65200-410700)
	High	4	12.8 (7.1-23.1)	36200 (19900-68000)
	Global			357100 (182800-684300)

* Estimates from meta-analyses.

† Based on all 54 studies. Of 52 studies, three studies for 5-9 years, 33 studies for 5-14 years, and 17 studies for 5-19 years.

‡ Based on data from a mixture of 5-14 years and 5-19 years.

§ Based on data for 5-19 years.

Appendix 4 Trend of hospital admission rates of ALRI

Data on ALRI hospital admission rates over four or more consecutive years are plotted in Figure S4.1. The trend of ALRI hospital admission rates varied by geographic locations: a decrease was found in Brazil (annual percent change, -3%), Canada (-3%), France (-3%), and Kenya (-8%); an increase in Denmark (+4%), Netherlands (+4%), and Taiwan, China (+14%). We did not found significant change of ALRI hospital admission rates over time in other locations.

We plotted multi-year studies with influenza-associated ALRI hospital admission rates for four or more consecutive years in Figure S4.2.

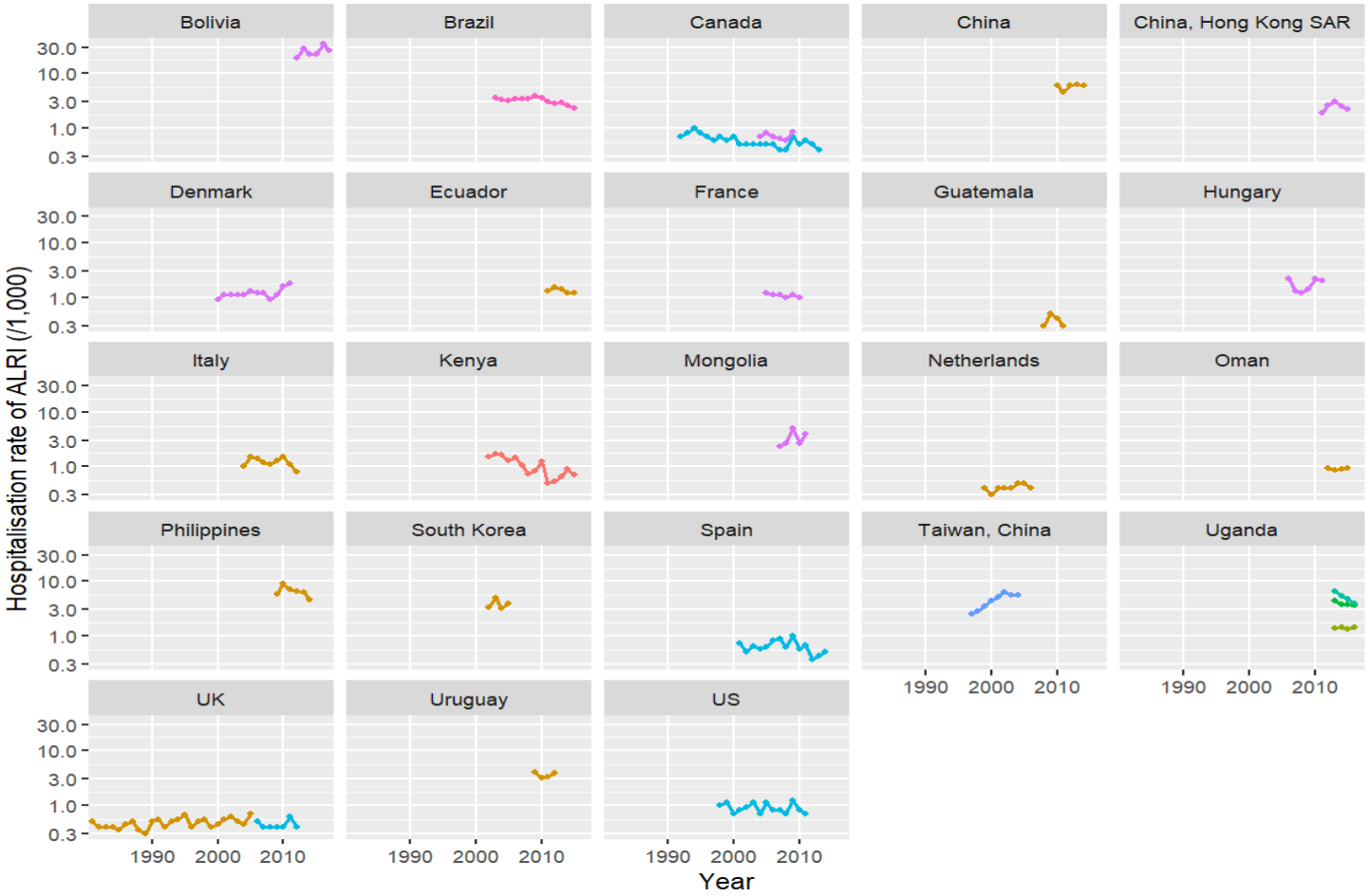


Figure S4.1 Trend of ALRI hospital admission rates in children and adolescents between 5 and 19 years.

For each region/country, the points represent the rates for each year, and the lines connecting these points represent the rates over years for one site or one age band. Different colours represent data from different sites or age bands. For example, in Canada the purple line represent the national rates for 5-19 years, and the blue line represent the rates among children and adolescents aged 5-17 years in Ontario, Canada.

Figure S4.2 Trend of influenza-associated ALRI hospital admission rates in children and adolescents between 5 and 19 years.

For each region/country, the points represent the rates for each year, and the lines connecting these points represent the rates over years for one site or one age band. Different colours represent data from different sites or age bands.

Appendix 5 Details of included studies

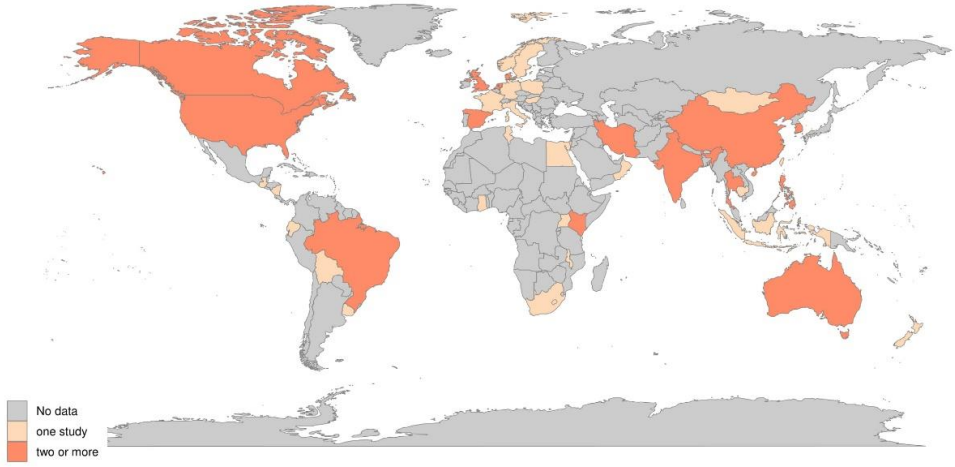


Figure S5.1 Countries of included all-cause ALRI hospital admission rate data

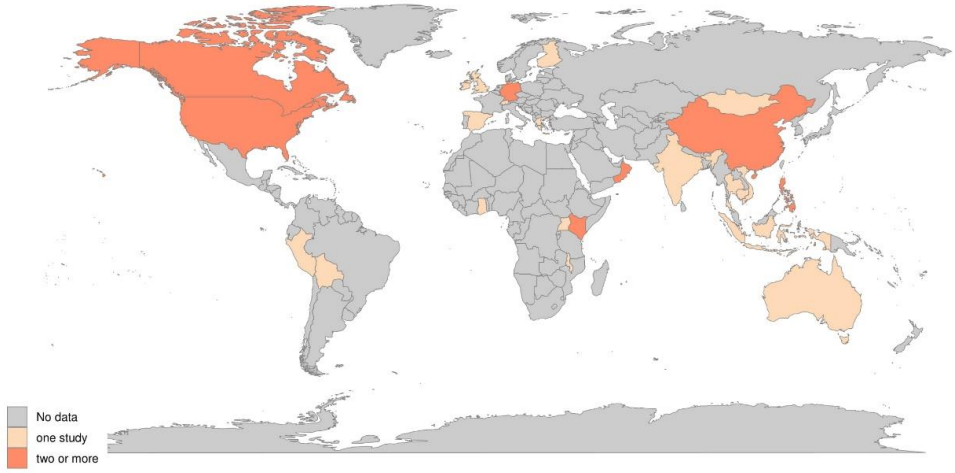


Figure S5.2 Countries of included influenza-associated ALRI hospital admission rate data

Table S5.1 Details of studies with data on hospital admission rates of ALRI per 1,000 children and adolescents per year.*

Study ID	Location (reference)	Case definition	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p1000	Baguio, Philippines (2010-2011)[7]	ALRI	NA	NA	NA	7.3	NA	NA
p1006	Baguio, Philippines (2012-2014) [8]	ALRI	NA	NA	NA	5.5	NA	NA
p1015	Chrzanów County, Poland (2006-2008) [9]	ALRI	NA	NA	NA	2.9	NA	NA
p113	Canada (2004-2010) [10]	ALRI	1.3	0.5	0.4	0.9	0.7	NA
p117	Edmonton, Canada (2000-2002) [11]	CXR-pneumonia	NA	NA	NA	NA	NA	NA
p125	multisites, Canada (1995-2001) [12]	ARI	NA	NA	NA	2.6	NA	NA
p1483	South Africa (2007-2012) [13]	ARI	NA	NA	NA	NA	4.3	NA
p1503	South Korea (2002-2005) [14]	ALRI	NA	NA	NA	3.9	NA	NA
p1505	South Korea (2007-2014) [15]	ALRI	NA	NA	NA	NA	NA	NA
p1516	Spain (1995-1998) [16]	ALRI	1.1	NA	NA	NA	NA	NA
p1517	Spain (2001-2014) [17]	ARI	1.2	NA	NA	NA	NA	0.6
p1522	Sweden (2005) [18]	ARI	1.1	NA	NA	NA	NA	NA
p1532	Sa Kaeo, Thailand (2002-2003) [19]	ARI	NA	NA	NA	3.3	NA	NA
p1541	Sa Kaeo and Nakhon Phanom, Thailand (2003-2007) [20]	ARI	NA	NA	NA	NA	2.7	NA
p1546	Sa Kaeo and Nakhon Phanom, Thailand (2008-2011) [21]	ARI	NA	NA	NA	NA	5.6	NA
p1548	Sfax, Tunisia (2003-2015) [22]	ALRI	0.4	0.2	NA	0.3	NA	NA
p1566	Wakiso, Uganda (2013-2016) [23]	ALRI	NA	NA	NA	2.4	NA	NA
p1571	England, UK (2001-2013) [24]	ALRI	1.6	0.7	NA	1.1	NA	NA
p1577	North East England, UK (2001-2002) [25]	CXR-pneumonia	NA	NA	NA	NA	NA	NA
p1578	North East England, UK (2008-2009) [26]	CXR-pneumonia	NA	NA	NA	NA	NA	NA
p1583	Scotland, UK (1981-2005) [27]	ALRI	0.6	0.3	NA	0.5	NA	NA
p1587	Scotland, UK (2000-2012) [28]	ALRI	NA	NA	NA	NA	NA	0.4
p1604	two municipalities, Uruguay (2009-2012) [29]	ALRI	NA	NA	NA	3.5	NA	NA
p1640	US (1996-2011) [30]	ARI	NA	NA	NA	NA	NA	0.9
p165	Ontario, Canada (1992-2014) [31]	ALRI	NA	NA	NA	NA	NA	0.6
p1654	US (2012-2014) [32]	ALRI	NA	NA	NA	NA	0.7	NA
p1871	Beijing, China (2014-2016) [33]	ARI AND fever	NA	NA	NA	3	NA	NA
p226	Beijing, China (2017-2018) [34]	ARI AND fever	NA	NA	NA	3.2	NA	NA
p286	China, Hong Kong SAR (2005) [35]	ALRI	NA	NA	NA	3.1	NA	NA

* ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. ARI OR fever: any respiratory symptoms OR fever requiring hospital admission. y: years.

Study ID	Location (reference)	Case definition	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p321	China, Hong Kong SAR (2011-2015) [36]	ALRI	NA	NA	NA	NA	2.4	NA
p355	Jing Zhou, China (2010-2012) [37]	ARI AND fever	36.9	6.3	NA	21	NA	NA
p416	Suzhou, China (2010-2014) [38]	ARI	NA	NA	NA	5.8	NA	NA
p420	Taiwan, China (1997-2004)[†] [39]	ALRI	NA	NA	NA	NA	NA	NA
p433	Denmark (1995-1999) [40]	ALRI	1.8	0.6	0.5	1.2	1	NA
p440	Denmark (1997-2011) [41]	ALRI	NA	NA	NA	NA	1.2	NA
p447	Ecuador (2011-2015) [42]	ARI	NA	NA	NA	1.3	NA	NA
p45	Mornington, Australia (2007-2011) [43]	ALRI	NA	NA	NA	2.6	NA	NA
p454	Damanhour, Egypt (2009-2012) [44]	ARI AND fever	NA	NA	NA	NA	0.9	NA
p509	Rhône-Alpes, France (2005-2010) [45]	ALRI	NA	NA	NA	NA	1.1	NA
p524	Kiel, Germany (1996-1999) [46]	ALRI	NA	NA	NA	NA	NA	NA
p532	Shai-Osudoku and Ningo-Prampram, Ghana (2013-2015) [47]	ARI AND fever	NA	NA	NA	2.3	NA	NA
p55	Western Australia (1996-2012) [48]	ALRI	2	NA	NA	NA	NA	NA
p557	Santa Rosa and Quetzaltenango, Guatemala (2007-2011) [49]	ARI	NA	NA	NA	0.4	NA	NA
p566	Hungary (2006-2011) [50]	ALRI	NA	NA	NA	NA	1.7	NA
p567	Ballabgarh, India (2010-2012) [51]	ARI	NA	NA	NA	4.3	NA	NA
p572	Ballabgarh, India (2012-2014) [52]	ALRI	2.1	NA	NA	NA	NA	NA
p62	Brazil (2003-2007 for hCFR; 2003/2004 for admission) [53]	ALRI	3.4	1.3	1.2	2.3	2	NA
p630	Wonosari, Indonesia (2013-2016) [54]	ARI AND fever	NA	NA	NA	1.5	NA	NA
p633	Iran (2015) [55]	ARI AND fever	NA	NA	NA	0.3	NA	NA
p646	multisites, Iran (2012-2013) [56]	ARI AND fever	NA	NA	NA	1.3	NA	NA
p669	Negev, Israel (1989-1991) [57]	ALRI	NA	NA	NA	1.4	NA	NA
p672	Veneto, Italy (2004-2012) [58]	ALRI	NA	NA	NA	1.2	NA	NA
p689	Bondo, Kenya (2001-2003) [59]	ALRI	0.8	0.8	2	0.8	1.1	NA
p693	Bondo, Kenya (2007-2009) [60]	ALRI	NA	NA	NA	NA	2.9	NA
p704	Dadaab, Kenya (2010-2012) [61]	ALRI	NA	NA	NA	1.5	NA	NA
p705	Karemo, Kenya (2009-2012) [62]	ARI	NA	NA	NA	NA	NA	2
p706	Kenya (2012-2014) [63]	ARI AND fever	NA	NA	NA	0.7	NA	NA
p73	Brazil (2005-2015) [64]	ALRI	3.1	NA	NA	NA	NA	1.8
p731	Kilifi, Kenya (2002-2015) [65]	ALRI	NA	NA	NA	NA	NA	NA
p748	Blantyre, Malawi (2011-2014) [66]	ALRI	0.1	0.1	NA	0.1	NA	NA
p774	multisites, Mongolia (2007-2012) [67]	ALRI	4.8	NA	NA	NA	3.3	NA
p877	Netherlands (2001-2007) [68]	ARI	NA	NA	NA	0.4	NA	NA

[†] Data were reported for other age bands, so are not presented in this table.

Study ID	Location (reference)	Case definition	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p882	Netherlands (2008-2011) ‡ [69]	ALRI	NA	NA	NA	NA	NA	NA
p939	Auckland, New Zealand (1993-1996) [70]	ARI	NA	NA	NA	1.4	NA	NA
p94	La Paz, Bolivia (2012-2017) [71]	ARI AND fever	NA	NA	NA	NA	25.4	NA
p945	Leo´n, Nicaragua (2008-2015) [72]	CXR-pneumonia	NA	NA	NA	0.8	NA	NA
p954	Norway (2008-2009) [73]	ALRI	NA	NA	NA	NA	1.2	NA
p983	Oman (2012-2015) [74]	ARI	NA	NA	NA	0.9	NA	NA
p99	Svay Rieng, Cambodia (2015-2016) § [75]	ALRI	NA	NA	NA	NA	NA	NA

‡ Data were reported for other age bands, so are not presented in this table.
§ Data were reported for other age bands, so are not presented in this table.

Table S5.2 Details of studies with data on incidence rates of ALRI (per 1,000 per year) *

Study ID	Location (reference)	Case definition	Age	Rate (/1,000)
P1595	UK (1991-2003)[76]	ALRI	5-14 y	0.8
p1597	UK (1995; 2000) [77]	ALRI	5-16 y	44.4
p1600	UK (1997-2012) [78]	ALRI	5-12 y	4.6
p1638	US (1994-2007) [79]	ALRI	6-18 y	10.8
p1657	US (2008-2014) [80]	ALRI	5-17 y	13.5
p1706	Washington State, US (1998-2004) [81]	ALRI	5-17 y	11.7
p572	Ballabgarh, India (2012-2014) [52]	ALRI	5-9 y	30.2
p839	Netherlands (1995-2001) [82]	ALRI (13% of influenza cases)	5-17 y	18.7
p877	Netherlands (2001-2007) [68]	ALRI	5-14 y	4.6

* ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

Table S5.3 Details of studies with data on in-hospital case-fatality ratios (hCFRs) of ALRI. *

Study ID	Location (reference)	Case definition	Cases for 5-9 y	hCFR for 5-9 y (%)	Cases for 5-14 y	hCFR for 5-14 y (%)	Cases for 5-19 y	hCFR for 5-19 y (%)
p1015	Chrzanów County, Poland (2006-2008) [9]	ALRI	NA	NA	269	0	NA	NA
p1071	multisites, South Africa (2009-2012)[83]	ALRI	NA	NA	577	2.1	NA	NA
p113	Canada (2004-2010) [10]	ALRI	10071	0.3	13274	0.5	17372	0.8
p1505	South Korea (2007-2014)† [15]	ALRI	NA	NA	NA	NA	NA	NA
p1513	Spain (1995-1996) [84]	ALRI	4303	0.3	NA	NA	NA	NA
p1527	Sa Kaeo (2002-2003) [85]	CXR-pneumonia	34	11.8	NA	NA	NA	NA
p1541	Sa Kaeo and Nakhon Phanom, Thailand (2003-2007) [20]	ARI	NA	NA	NA	NA	1069	0.4
p1548	Sfax, Tunisia (2003-2015) [22]	ALRI	369	0.3	562	0.9	NA	NA
p1565	multisites, Uganda (2010-2015) [86]	ARI	NA	NA	139	1.4	NA	NA
p1568	England, UK (2000-2007) [87]	ARI	NA	NA	52523	0.1	NA	NA
p1602	Paysandú and Salto, Uruguay (2001-2004) [88]	ALRI	NA	NA	361	1.4	NA	NA
p1654	US (2012-2014) [32]	ALRI	NA	NA	NA	NA	68483	0.7
p246	Guangzhou, China (2009-2012)‡ [89]	ALRI	NA	NA	NA	NA	NA	NA
p321	China, Hong Kong SAR (2011-2015) [36]	ALRI	NA	NA	NA	NA	11337	0.9
p420	Taiwan, China (1997-2004) [39]	ALRI	2562	0.1	3015	0.1	NA	NA
p440	Denmark (1997-2011) [41]	ALRI	NA	NA	NA	NA	11725	0.4
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	693	0.7	1049	0.8	1366	1
p566	Hungary (2006-2011) [50]	ALRI	NA	NA	NA	NA	16473	0.5
p62	Brazil (2003-2007 for hCFR; 2003/2004 for Hos) [53]	ALRI	293030	0.3	405175	0.3	512235	0.5
p633	Iran (2015) [55]	ARI AND fever	NA	NA	3329	3	NA	NA
p689	Bondo, Kenya (2001-2003) [59]	ALRI	85	5.9	135	5.2	248	6
p739	multisites, Kenya (2014-2018) [91]	ALRI	1462	6.4	1825	7.9	NA	NA

* ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

† Data were reported for other age bands, so are not presented in this table.

‡ Data were reported for other age bands, so are not presented in this table.

Study ID	Location (reference)	Case definition	Cases for 5-9 y	hCFR for 5-9 y (%)	Cases for 5-14 y	hCFR for 5-14 y (%)	Cases for 5-19 y	hCFR for 5-19 y (%)
p780	Maputo, Mozambique (2014-2016) [92]	ARI AND fever	NA	NA	294	0	NA	NA
p794	eight African countries (2009-2012) [93]	ARI AND fever	NA	NA	NA	NA	NA	NA
p90	Rio de Janeiro, Brazil (1996-2011) [§] [94]	ALRI	NA	NA	NA	NA	NA	NA
p983	Oman (2012-2015) [74]	ARI	NA	NA	1837	0.8	NA	NA
p992	Asuncion, Paraguay (2004-2013) [95]	CXR-pneumonia	NA	NA	165	10.9	NA	NA

[§] Data were reported for other age bands, so are not presented in this table.

Table S5.4 Details of studies with data on hospital admission rates of influenza-associated ALRI (per 1,000 children and adolescents per year). [*†]

Study ID	Location (reference)	Case definition	Specimen and diagnostic test	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p1000	Baguio, Philippines (2010-2011) [7]	ALRI	NPS and OPS; PCR	NA	NA	NA	1.5	NA	NA
p1006	Baguio, Philippines (2012-2014) [8]	ALRI	NPS and OPS; PCR	NA	NA	NA	1.5	NA	NA
p105	Canada (2003-2008) [96]	ICD-coded lab-confirmed influenza	NA; NA	0.1	0	0	0	0	NA
p1070	Singapore (2013-2014) [97]	ARI AND fever	NS; DFA	1.2	NA	NA	NA	0.5	NA
p108	Canada (2003-2007; 2010-2014)[‡] [98]	ARI	NA; NA	NA	NA	NA	NA	NA	NA
p1519	Spain (2010-2016) [99]	ALRI	NA; NA	NA	NA	NA	0	NA	NA
p1525	Basel, Switzerland (2001-2002) [§] [100]	ARI	NPS; PCR and rapid antigen test	NA	NA	NA	NA	NA	NA
p1545	Sa Kaeo and Nakhon Phanom, Thailand (2005-2008) [101]	ARI	NPS; mainly PCR (and culture)	0.6	0.4	0.1	0.5	0.3	NA
p1566	Wakiso, Uganda (2013-2016) [23]	ALRI	NPS and OPS; PCR	NA	NA	NA	0.3	NA	NA
p1574	England, UK (2010-2015) [102]	ARI AND fever	NA; NA	NA	NA	NA	0.1	NA	NA
p1610	Colorado, US (2004-2008) [103]	ARI	NA; PCR, DFA, and rapid test	NA	NA	NA	NA	NA	0.1
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	0.1	NA	NA	NA	NA	0.1
p1618	US (1996-1998) [**] [105]	ALRI	mostly NPS; PCR, culture and EIA (enzyme immunoassay)	NA	NA	NA	NA	NA	NA
p1625	Philadelphia, US (2000-2004) [106]	ARI	Nasal aspirate; culture, DFA, and solid-phase immunoassay (SPIA)	NA	NA	NA	NA	NA	0.2
p1637	Salt Lake County, Utah, US (2001-2004) [107]	ARI	NPA; culture and DFA	NA	NA	NA	0.1	NA	NA

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash.

[‡] Data were reported for other age bands, so are not presented in this table.

[§] Data were reported for other age bands, so are not presented in this table.

[**] Data were reported for other age bands, so are not presented in this table.

Study ID	Location (reference)	Case definition	Specimen and diagnostic test	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p1656	US (2003-2008) [108]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	NA	0.1
p1668	US (2008-2011) [109]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	0.2	NA
p1671	US (2012-2013) [110]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	0.3	NA
p1672	US (2013-2014) [111]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	0.2	NA
p1686	US (2017-2018) [112]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	NA	0.4
p1690	US (2018-2019) [113]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	NA	0.4
p1700	Utah, US (2016-2017) [114]	ARI	NA; PCR (sensitivity adjusted)	NA	NA	NA	NA	NA	0.3
p1735	Nha Trang, Viet Nam (2007-2012) [115]	ALRI	NPS; PCR	NA	NA	NA	0.2	NA	NA
p1871	Beijing, China (2014-2016) [33]	ARI AND fever	TS; PCR	NA	NA	NA	0.6	NA	NA
p226	Beijing, China (2017-2018) [34]	ARI AND fever	pharyngeal swab; PCR	NA	NA	NA	0.8	NA	NA
p328	China, Hong Kong SAR (2003-2006) [116]	ARI AND fever	NPA; DFA and culture	2.7	0.5	NA	1.5	NA	1.3
p330	China, Hong Kong SAR (2005-2011) [117]	ICD-coded lab-confirmed influenza	NPA; Immunofluorescence and culture	2.7	0.6	NA	1.6	NA	1.1
p355	Jing Zhou, China (2010-2012) [37]	ARI AND fever	NPA; PCR and others	6.8	1	NA	3.8	NA	NA
p39	Australia (2006-2015) [118]	ICD-coded lab-confirmed influenza	NA; PCR and others	NA	NA	NA	NA	NA	0.1
p491	Turku, Finland (1988-2004) [119]	ARI	NPA; time-resolved fluoroimmunoassay and rapid influenza test	0.2	0.1	NA	0.2	NA	NA
p520	Germany (2004-2010) [120]	ICD-coded lab-confirmed influenza	NA; NA	NA	NA	NA	0.1	NA	NA
p525	Kiel, Germany (1996-2001) [††] [121]	ALRI	NPA; PCR	NA	NA	NA	NA	NA	NA
p532	Shai-Osudoku and Ningo-Prampram, Ghana (2013-2015) [47]	ARI AND fever	NPS and OPS; PCR	NA	NA	NA	0.3	NA	NA
p536	Athens, Greece (2002-2005) [122]	ARI OR fever	NPA; PCR	1	NA	NA	NA	NA	NA
p567	Ballabgarh, India (2010-2012) [51]	ARI	NS and TS; PCR	NA	NA	NA	1.5	NA	NA
p630	Wonosari, Indonesia (2013-2016) [54]	ARI AND fever	NA; PCR	NA	NA	NA	0.3	NA	NA

[††] Data were reported for other age bands, so are not presented in this table.

Study ID	Location (reference)	Case definition	Specimen and diagnostic test	5-9 y	10-14 y	15-19 y	5-14 y	5-19 y	5-17 y
p655	Ireland (2010-2011) [123]	ARI AND fever	NA; PCR	1.7	1.3	NA	1.5	NA	NA
p693	Bondo, Kenya (2007-2009) [60]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	0.2	NA
p704	Dadaab, Kenya (2010-2012) [61]	ALRI	NPS and OPS; PCR	NA	NA	NA	0.7	NA	NA
p705	Karemo, Kenya (2009-2012) [62]	ARI	NPS and OPS; PCR	NA	NA	NA	NA	NA	0.2
p738	Kilifi, Kenya (2007-2010) ‡‡ [124]	ALRI	NPW or NPA; PCR	NA	NA	NA	NA	NA	NA
p748	Blantyre, Malawi (2011-2014) [66]	ALRI	NPA; PCR	NA	NA	NA	0	NA	NA
p774	multisites, Mongolia (2007-2012) [67]	ALRI	NPS; PCR and culture	0.9	NA	NA	NA	1	NA
p94	La Paz, Bolivia (2012-2017) [71]	ARI AND fever	NA; PCR (flu) and IFA (respiratory syncytial virus)	NA	NA	NA	NA	4.8	NA
p963	Sohar, Oman (2010-2013) [125]	ARI	NPS and OPS; PCR	NA	NA	NA	0	NA	NA
p983	Oman (2012-2015) [74]	ARI	NA; PCR	NA	NA	NA	0.3	NA	NA
p99	Svay Rieng, Cambodia (2015-2016) §§ [75]	ALRI	NPA; PCR	NA	NA	NA	NA	NA	NA
p995	multisites, Peru (2009-2015) [126]	ARI AND fever	NPS and OPS; PCR	NA	NA	NA	NA	NA	0.2

‡‡ Data were reported for other age bands, so are not presented in this table.
§§ Data were reported for other age bands, so are not presented in this table.

Table S5.5 Details of all studies with data on proportion positives of influenza in hospitalised ALRI. [*†‡]

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p1000	Baguio, Philippines (2010-2011)[127]	ALRI	NPS and OPS; PCR	NA	NA	196	14.8	NA	NA
p1006	Baguio, Philippines (2012-2014)[8]	ALRI	NPS and OPS; PCR	NA	NA	268	20.1	NA	NA
p101	Yaounde, Cameroon (2011-2013) [128]	ARI AND fever	NPS; PCR	NA	NA	40	10	NA	NA
p1018	Riyadh, Saudi Arabia (2005-2010) [129]	ALRI	NPA; DFA	NA	NA	NA	NA	50	0
p1071	multisites, South Africa (2009-2012) [83]	ALRI	NPS and TS; PCR	NA	NA	560	11.4	NA	NA
p1488	Seongnam, South Korea (2006-2016) [130]	ALRI	Nasal aspirate; PCR	1019	11.5	NA	NA	1723	11.2
p1489	Seoul, South Korea (1996-1998) [131]	ALRI	NPA; Culture	100	10	250	5.2	NA	NA
p1492	Seoul, South Korea (2006-2007) [132]	ALRI	NP aspiration; PCR	63	9.5	75	8	NA	NA
p1520	multisies, Spain (2011-2013) [133]	ALRI	NP specimen; PCR	NA	NA	40	25	NA	NA
p1554	Bursa, Turkey (2015-2018) [134]	ARI	NPS or BAL (bronchoalveolar lavage); PCR	NA	NA	NA	NA	84	22.6
p1557	Istanbul, Turkey (2010-2011) [135]	ARI	NPA; PCR	11	18.2	NA	NA	NA	NA
p1565	multisites, Uganda (2010-2015) [86]	ARI	NPS and OPS; PCR	NA	NA	404	13.9	NA	NA
P1566	multisites, Uganda (2013-2016)	ALRI	NPS and OPS; PCR	NA	NA	8912	13.0	NA	NA
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012)[136]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	408	9.1	NA	NA	683	9.8

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-11 years were re-classified as 5-9 years; 5-15 years and 6-16 years were re-classified as 5-14 years; 5-17 years, 5-18 years, 6-17 years, 7-17 years, and 6-18 years were re-classified as 5-19 years.

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p1733	Ho Chi Minh, Vietnam (2004-2008) [137]	ARI	NS, TS, and NPA; PCR	NA	NA	13	30.8	NA	NA
p1739	Vietnam (2011-2014) [138]	ALRI	TS and tracheal aspirate; PCR	NA	NA	NA	NA	156	49.0
p1870	Cairo, Egypt (2010-2014) [139]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	191	3.7
P1871	Beijing, China (2014-2016) [33]	ARI AND fever	TS; PCR	NA	NA	12	50.0	NA	NA
p1873	Lanzhou, China (2004-2005) [140]	CXR-pneumonia	NPA; DFA	NA	NA	133	12.8	NA	NA
p1901	Chengdu, China (2007) [141]	ALRI	NP secretion; DFA	NA	NA	121	9.1	NA	NA
p1931	Changsha, China (2015) [142]	ARI	NP secretion; DFA	NA	NA	285	1.1	NA	NA
p216	Beijing and Shandong, China (2012-2015) [143]	ALRI	NPS, NPA and sputum; PCR	NA	NA	850	5.2	NA	NA
p221	Beijing, China (2007-2010) [144]	ALRI	NP secretion; PCR	NA	NA	358	5.9	NA	NA
p226	Beijing, China (2017-2018)[34]	ARI AND fever	pharyngeal swab; PCR	NA	NA	64	23.4	NA	NA
p241	Beijing, China (2011-2012) [145]	ALRI	Tracheal aspirate; PCR	NA	NA	NA	NA	93	5.4
p264	Changsha, China (2007-2008) [146]	ALRI	NPA; PCR	NA	NA	42	7.1	NA	NA
p311	Hebei, China (2007-2008) [147]	ALRI	TS; DFA	NA	NA	361	7.5	NA	NA
p331	China, Hong Kong SAR (2009-2013) [148]	ARI AND fever	NA; DFA, culture, PCR	NA	NA	NA	NA	1037	17.9
p367	Lanzhou,China (2006-2009) [149]	ALRI	NA; PCR	NA	NA	67	7.5	NA	NA
p376	multisites,China (2009-2013) [150]	ALRI	NS or aspirate, sputum, BAL or lung puncture; PCR	2618	8.5	3510	7.9	NA	NA
p386	Shanghai, China (2003-2006) [151]	ALRI	NPA; DFA	NA	NA	974	0.8	NA	NA
p387	Shanghai,China (2013-2015) [152]	ALRI	NP secretion; DFA	NA	NA	NA	NA	1164	1.8
p388	Shanghai, China (2016-2017) [153]	ALRI	NPS or sputum specimen; PCR	NA	NA	106	17.9	NA	NA
p396	Shantou, Shenzhen and Jieyang, China (2007-2011) [154]	ARI	NPS or sputum specimen; PCR	NA	NA	112	23.2	NA	NA
p399	Shantou, Shenzhen and Jieyang, China (2006-2008) [155]	ALRI	PCR; PCR	NA	NA	30	6.7	NA	NA
p400	nine cities, North China (2015) [156]	ALRI	NP secretion; PCR	NA	NA	331	8.2	NA	NA
p404	Shenzhen, China (2012-2015) [157]	ARI AND fever	NS; DFA	NA	NA	1568	1	NA	NA
p421	Tianjin, China (2015-2016) [158]	ARI AND fever	TS; PCR	NA	NA	62	4.8	NA	NA

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p427	Zhejiang, China (2011-2015) [159]	ALRI	NPS and TS; PCR	NA	NA	501	10	NA	NA
p428	multisites, DR Congo (2015) [160]	ARI AND fever	TS and NS; PCR	266	10.2	NA	NA	490	8.2
p452	Damanhour, Egypt (2013) [161]	ARI AND fever	NPS and OPS; PCR	NA	NA	146	26	NA	NA
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	OPS and NPS; PCR	796	19	NA	NA	1661	18.8
p476	multisites, Egypt (2012-2015) [162]	ARI AND fever	NPS and OPS; PCR	NA	NA	980	16.2	NA	NA
p52	Perth, Australia (2000-2005) [163]	ALRI	NPA; PCR	102	1	NA	NA	NA	NA
p523	Kiel, Germany (1995-1999) [164]	ALRI	NPA; PCR	NA	NA	107	23.4	NA	NA
p532	Shai-Osudoku and Ningo-Prampram, Ghana (2013-2015)[47]	ARI AND fever	NPS and OPS; PCR	NA	NA	108	13	NA	NA
p540	Athens, Greece (pre 2003) [165]	CXR-pneumonia	NPW; PCR	NA	NA	75	6.7	NA	NA
p612	Odisha, India (2012-2014) [166]	ARI	multiple; PCR	82	4.9	104	3.8	NA	NA
p619	East Jakarta, Indonesia (2011-2012) [167]	ARI AND fever	NS and TS; PCR	NA	NA	NA	NA	306	17
p624	East Jakarta, Indonesia (2011-2014) [168]	ARI AND fever	NS and TS; PCR	NA	NA	NA	NA	614	18.1
p630	multisites, Indonesia (2013-2016)[54]	ARI AND fever	NA; PCR	NA	NA	284	20.4	NA	NA
p671	Naples, Italy (2016-2017) [169]	ARI	NPS; PCR	36	13.9	NA	NA	NA	NA
p679	multisites, Jordan (2008-2014) [170]	ALRI	NPS and OPS; PCR	NA	NA	252	13	NA	NA
p693	Bondo, Kenya (2007-2009)[60]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	204	59.3
p704	Dadaab, Kenya (2010-2012)[61]	ALRI	NPS and OPS; PCR	NA	NA	180	41.1	NA	NA
p705	Karemo, Kenya (2009-2012)[62]	ARI	NPS and OPS; PCR	NA	NA	NA	NA	126	10.3
p738	Kilifi, Kenya (2007-2010)[124]	ALRI	NPW or NPA; PCR	NA	NA	149	8.1	NA	NA
p744	Beirut suburbs and the southern province of Lebanon (2015-2016)[171]	ARI AND fever	NPS; PCR	NA	NA	68	16.2	NA	NA
p748	Blantyre, Malawi (2011-2014)[66]	ALRI	NPA; PCR	NA	NA	82	23.2	NA	NA
p744	Southern province, Lebanon (2015-2016) [171]	ARI AND fever	Nasopharyngeal swabs; PCR	NA	NA	68	16.2	NA	NA
p775	Morocco (2007-2009) [172]	ALRI	NPS and OPS; IFA	NA	NA	65	6.2	NA	NA

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p780	Maputo, Mozambique (2014-2016) [92]	ARI AND fever	NPS and OPS; PCR	NA	NA	294	4.8	NA	NA
p781	11 countries AFR (2010-2012) [173]	ALRI	NP and OP specimen; PCR	NA	NA	200	10	NA	NA
p784	14 countries AFR and EMR (2006-2010) [174]	ALRI	NA; PCR	3044	11	4117	11	NA	NA
p829	Multi-country (2008-2009) [175]	ARI AND fever	multiple; PCR	NA	NA	126	15.9	NA	NA
p94	La Paz, Bolivia (2012-2017)[71]	ARI AND fever	NA; PCR (flu) and IFA (RSV)	NA	NA	NA	NA	261	17.6
p953	Lørenskog, Norway (2012-2014) [176]	CXR-pneumonia	NA; NA	NA	NA	NA	NA	40	7.5
p983	Oman (2012-2015)[177]	ARI	NA; PCR	NA	NA	1837	29.9	NA	NA
p99	Svay Rieng, Cambodia (2015-2016)[75]	ALRI	NPA; PCR	NA	NA	251	23.1	NA	NA

Table S5.6 Details of studies with data on hospital admission rates of respiratory syncytial virus-associated ALRI (per 1,000 children and adolescents per year). [*†‡]

Study ID	Location (reference)	Case definition	Specimen and test	5-9 y	5-14 y	5-19 y
p1000	Baguio, Philippines (2010-2011) [7]	ALRI	NPS and OPS; PCR	NA	0.5	NA
p1006	Baguio, Philippines (2012-2014) [8]	ALRI	NPS and OPS; PCR	NA	0.3	NA
p108	Canada (2003-2007; 2010-2014) [98]	ARI	NA; NA	NA	0	NA
p1525	Basel, Switzerland (2001-2002) [100]	ARI	NPS; PCR and rapid antigen test	NA	NA	NA
p1541	Sa Kaeo and Nakhon Phanom, Thailand (2003-2007) [20]	ARI	NPS; PCR	NA	NA	0.1
p1546	Sa Kaeo and Nakhon Phanom, Thailand (2008-2011) [21]	ARI	NPS; PCR	NA	NA	0.2
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	0.1	NA	0
p1618	US (1996-1998) [105]	ALRI	mostly NPS; PCR, culture and EIA (enzyme immunoassay)	NA	NA	0.1
p1621	multisies, US (2001-2012) [178]	ALRI	NA; NA	0.1	NA	0
p31	Australia (2006-2015) [179]	ALRI	NA; NA	NA	0.1	NA
p328	China, Hong Kong SAR (2003-2006) [116]	ARI AND fever	NPA; DFA and culture	0.2	0.1	0.1
p454	Damanhour, Egypt (2009-2012) [44]	ARI AND fever	NPS and OPS; PCR	NA	NA	0
p525	Kiel, Germany (1996-2001) [121]	ALRI	NPA; PCR	NA	0	NA
p562	Santa Rosa and Quetzaltenango, Guatemala (2008-2012) [180]	ARI	NPS and OPS; PCR	NA	NA	0.1
p705	Karemo, Kenya (2009-2012) [62]	ARI	NPS and OPS; PCR	NA	NA	0.1
p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	0	NA
p748	Blantyre, Malawi (2011-2014) [66]	ALRI	NPA; PCR	NA	0	NA
p94	La Paz, Bolivia (2012-2017) [71]	ARI AND fever	NA; PCR (flu) and IFA (respiratory syncytial virus)	NA	NA	0.5
p957	Trondheim, Norway (2006-2015) [181]	ALRI	NPA; PCR	NA	0.1	NA

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-15 years and 5-16 years were re-classified as 5-14 years; 5-17 years and 5-18 years and 6-18 years were re-classified as 5-19 years.

Table S5.7 Details of all studies with data on proportion positives of respiratory syncytial virus in hospitalised ALRI. [*†‡]

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p1000	Baguio, Philippines (2010-2011) [7]	ALRI	NPS and OPS; PCR	NA	NA	266	6.8	NA	NA
p1006	Baguio, Philippines (2012-2014) [8]	ALRI	NPS and OPS; PCR	NA	NA	268	4.5	NA	NA
p101	Yaounde, Cameroon (2011-2013) [128]	ARI AND fever	NPS; PCR	NA	NA	40	7.5	NA	NA
p1018	Riyadh, Saudi Arabia (2005-2010) [129]	ALRI	NPA; DFA	NA	NA	NA	NA	50	28.0
p1071	multisites, South Africa (2009-2012) [83]	ALRI	NPS and TS; PCR	NA	NA	550	6.5	NA	NA
p1488	Seongnam, South Korea (2006-2016) [130]	ALRI	Nasal aspirate; PCR	NA	NA	NA	NA	1723	10
p1489	Seoul, South Korea (1996-1998) [131]	ALRI	NPA; Culture	NA	NA	250	0	NA	NA
p1492	Seoul, South Korea (2006-2007) [132]	ALRI	NP aspiration; PCR	NA	NA	75	1.3	NA	NA
p1520	multisies, Spain (2011-2013) [133]	ALRI	NP specimen; PCR	NA	NA	40	10	NA	NA
p1541	Sa Kaeo and Nakhon Phanom, Thailand (2003-2007) [20]	ARI	NPS; PCR	NA	NA	NA	NA	1069	5.1
p1546	Sa Kaeo and Nakhon Phanom, Thailand (2008-2011) [21]	ARI	NPS; PCR	NA	NA	NA	NA	1802	4.1
p1554	Bursa, Turkey (2015-2018) [134]	ARI	NPS or BAL (bronchoalveolar lavage); PCR	NA	NA	NA	NA	84	7.1
p1557	Istanbul, Turkey (2010-2011) [135]	ARI	NPA; PCR	11	18.2	NA	NA	NA	NA
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	NA	NA	NA	NA	683	7.6
p1733	Ho Chi Minh, Vietnam (2004-2008) [137]	ARI	NS, TS, and NPA; PCR	NA	NA	13	7.7	NA	NA

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-11 years were re-classified as 5-9 years; 5-15 years, 5-16 years, 6-12 years, 6-15 years, and 6-16 years were classified as 5-14 years; 5-17 years, 5-18 years, 6-17 years, 7-17 years, and 6-18 years were re-classified as 5-19 years.

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p1870	Cairo, Egypt (2010-2014) [139]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	191	18.8
P1872	San Luis Potosı, Mexico (2009-2010) [182]	ALRI	pharyngeal swab/NPS; PCR	NA	NA	NA	NA	62	9.7
p1873	Lanzhou, China (2004-2005) [140]	CXR-pneumonia	NPA; DFA	NA	NA	133	12.8	NA	NA
p1901	Chengdu, China (2007) [141]	ALRI	NP secretion; DFA	NA	NA	121	2.5	NA	NA
p1931	Changsha, China (2015) [142]	ARI	NP secretion; DFA	NA	NA	285	2.1	NA	NA
P1939	Jieyang, China (2016-2017) [183]	ALRI	NP secretion; DFA	NA	NA	67	1.5		
p216	Beijing and Shandong, China (2012-2015) [143]	ALRI	NPS, NPA and sputum; PCR	NA	NA	850	4.7	NA	NA
p221	Beijing, China (2007-2010) [144]	ALRI	NP secretion; PCR	NA	NA	358	4.5	NA	NA
p241	Beijing, China (2011-2012) [145]	ALRI	Tracheal aspirate; PCR	NA	NA	NA	NA	93	47.3
p245	Chongqing, China (2006-2008) [184]	ALRI	NPA; PCR	NA	NA	313	0	NA	NA
p264	Changsha, China (2007-2008) [146]	ALRI	NPA; PCR	NA	NA	42	11.9	NA	NA
p311	Hebei, China (2007-2008) [147]	ALRI	TS; DFA	NA	NA	361	1.1	NA	NA
p367	Lanzhou,China (2006-2009) [149]	ALRI	NA; PCR	NA	NA	67	19.4	NA	NA
p376	multisites,China (2009-2013) [150]	ALRI	NS or aspirate, sputum, BAL or lung puncture; PCR	NA	NA	3510	4.3	NA	NA
p386	Shanghai, China (2003-2006) [151]	ALRI	NPA; DFA	NA	NA	974	3.6	NA	NA
p387	Shanghai,China (2013-2015) [152]	ALRI	NP secretion; DFA	NA	NA	NA	NA	1164	1.5
p388	Shanghai, China (2016-2017) [153]	ALRI	NPS or sputum specimen; PCR	NA	NA	106	2.8	NA	NA
p396	Shantou, Shenzhen and JIeyang, China (2007-2011) [154]	ARI	NPS or sputum specimen; PCR	NA	NA	112	13.4	NA	NA
p399	Shantou, Shenzhen and JIeyang, China (2006-2008) [155]	ALRI	PCR; PCR	NA	NA	30	0	NA	NA
p400	nine cities, North China (2015) [156]	ALRI	NP secretion; PCR	NA	NA	331	5.1	NA	NA
p404	Shenzhen, China (2012-2015) [157]	ARI AND fever	NS; DFA	NA	NA	1568	0.6	NA	NA
p421	Tianjin, China (2015-2016) [158]	ARI AND fever	TS; PCR	NA	NA	62	1.6	NA	NA
p454	Damanhour, Egypt (2009-2012) [44]	ARI AND fever	NPS and OPS; PCR	NA	NA	NA	NA	564	0.5
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	OPS and NPS; PCR	NA	NA	NA	NA	1661	4.8
p476	multisites, Egypt (2012-2015) [162]	ARI AND fever	NPS and OPS; PCR	NA	NA	980	4.2	NA	NA
p52	Perth, Australia (2000-2005) [163]	ALRI	NPA; PCR	103	12.6	NA	NA	NA	NA
p523	Kiel, Germany (1995-1999) [164]	ALRI	NPA; PCR	NA	NA	162	1.2	NA	NA
p524	Germany (1996-1999) [46]	ALRI	NPA; PCR	NA	NA	262	0.8	NA	NA

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p540	Athens, Greece (pre 2003) [165]	CXR-pneumonia	NPW; PCR	NA	NA	75	2.7	NA	NA
p562	Guatemala (2008-2012) [180]	ARI	NPS and OPS; PCR	NA	NA	NA	NA	403	7.7
p587	Lucknow, India (2011-2013) [185]	ARI AND fever	NPA; PCR	NA	NA	59	0	NA	NA
p612	Odisha, India (2012-2014) [166]	ARI	multiple; PCR	NA	NA	104	0	NA	NA
p671	Naples, Italy (2016-2017) [169]	ARI	NPS; PCR	36	2.8	NA	NA	NA	NA
p705	Karemo, Kenya (2009-2012) [62]	ARI	NPS and OPS; PCR	NA	NA	NA	NA	126	5.6
p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	NA	149	5.4	NA	NA
p742	Nairobi, Kenya, Slum (2007-2011) [186]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	1792	6.5
p829	Multi-country (2008-2009) [175]	ARI AND fever	multiple; PCR	NA	NA	125	2.4	NA	NA
p94	La Paz, Bolivia (2012-2017) [71]	ARI AND fever	NA; PCR (flu) and IFA (respiratory syncytial virus)	NA	NA	NA	NA	261	2.7
p953	Lørenskog, Norway (2012-2014) [176]	CXR-pneumonia	NA; NA	NA	NA	NA	NA	40	7.5

Table S5.8 Details of all studies with data on proportion positives of human metapneumovirus in hospitalised ALRI. [*†‡]

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p101	Yaounde, Cameroon (2011-2013) [128]	ARI AND fever	NPS; PCR	NA	NA	40	7.5	NA	NA
p1071	multisites, South Africa (2009-2012) [83]	ALRI	NPS and TS; PCR	NA	NA	550	2.4	NA	NA
p1488	Seongnam, South Korea (2006-2016) [130]	ALRI	Nasal aspirate; PCR	NA	NA	NA	NA	1723	0.9
p1492	Seoul, South Korea (2006-2007) [132]	ALRI	NPA; PCR	NA	NA	75	5.3	NA	NA
p1520	multisies, Spain (2011-2013) [133]	ALRI	NP specimen; PCR	NA	NA	40	5	NA	NA
p1557	Istanbul, Turkey (2010-2011) [135]	ARI	NPA; PCR	11	0	NA	NA	NA	NA
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	NA	NA	NA	NA	683	7.6
p1733	Ho Chi Minh, Vietnam (2004-2008) [137]	ARI	NS, TS, and NPA; PCR	NA	NA	13	7.7	NA	NA
p216	Beijing and Shandong, China (2012-2015) [143]	ALRI	NPS, NPA and sputum; PCR	NA	NA	850	1.8	NA	NA
p221	Beijing, China (2007-2010) [144]	ALRI	NP secretion; PCR	NA	NA	358	0.3	NA	NA
p241	Beijing, China (2011-2012) [145]	ALRI	Tracheal aspirate; PCR	NA	NA	NA	NA	93	2.2
p245	Chongqing, China (2006-2008) [184]	ALRI	NPA; PCR	NA	NA	227	0.9	NA	NA
p257	Guangzhou, China (2013-2016) [187]	ARI	TS; PCR	NA	NA	734	0.7	NA	NA
p264	Changsha, China (2007-2008) [146]	ALRI	NPA; PCR	NA	NA	42	11.9	NA	NA
p367	Lanzhou,China (2006-2009) [149]	ALRI	NA; PCR	NA	NA	67	6	NA	NA
p376	multisites,China (2009-2013) [150]	ALRI	NS, NA, sputum, BAL or lung puncture; PCR	NA	NA	3510	1	NA	NA
p387	Shanghai,China (2013-2015) [152]	ALRI	NP secretion; DFA	NA	NA	NA	NA	1164	0.1

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-11 years were re-classified as 5-9 years; 5-15 years, 5-16 years, 6-12 years, 6-15 years, and 6-16 years were classified as 5-14 years; 5-17 years, 5-18 years, 6-17 years, 7-17 years, and 6-18 years were re-classified as 5-19 years.

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p388	Shanghai, China (2016-2017) [153]	ALRI	NPS or sputum specimen; PCR	NA	NA	106	0.9	NA	NA
p396	Shantou, Shenzhen and JIeyang, China (2007-2011) [154]	ARI	NPS or sputum specimen; PCR	NA	NA	112	0	NA	NA
p399	Shantou, Shenzhen and JIeyang, China (2006-2008) [155]	ALRI	PCR; PCR	NA	NA	30	6.7	NA	NA
p400	nine cities, North China (2015) [156]	ALRI	NP secretion; PCR	NA	NA	331	0	NA	NA
p407	Suzhou, China (2005-2006) [188]	ALRI	Nasal aspirate; PCR	NA	NA	203	3	NA	NA
p411	Suzhou, China (2006-2009) [189]	ARI	Nasal aspirate; PCR	646	5.9	NA	NA	NA	NA
p421	Tianjin, China (2015-2016) [158]	ARI AND fever	TS; PCR	NA	NA	62	0	NA	NA
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	OPS and NPS; PCR	NA	NA	NA	NA	361	8.3
p540	Athens, Greece (pre 2003) [165]	CXR-pneumonia	NPW; PCR	NA	NA	75	1.3	NA	NA
p587	Lucknow, India (2011-2013) [185]	ARI AND fever	NPA; PCR	NA	NA	59	5.1	NA	NA
p612	Odisha, India (2012-2014) [166]	ARI	multiple; PCR	NA	NA	104	1	NA	NA
p671	Naples, Italy (2016-2017) [169]	ARI	NPS; PCR	36	2.8	NA	NA	NA	NA
p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	NA	149	4	NA	NA
p829	Multi-country (2008-2009) [175]	ARI AND fever	multiple; PCR	NA	NA	125	0.8	NA	NA
p953	Lørenskog, Norway (2012-2014) [176]	CXR-pneumonia	NA; NA	NA	NA	NA	NA	40	5

Table S5.9 Details of all studies with data on proportion positives of human parainfluenza virus in hospitalised ALRI. *†‡

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p101	Yaounde, Cameroon (2011-2013) [128]	ARI AND fever	NPS; PCR	NA	NA	40	5	NA	NA
p1018	Riyadh, Saudi Arabia (2005-2010) [129]	ALRI	NPA; DFA	NA	NA	NA	NA	50	0
p1071	multisites, South Africa (2009-2012) [83]	ALRI	NPS and TS; PCR	NA	NA	550	3.3	NA	NA
p1488	Seongnam, South Korea (2006-2016) [130]	ALRI	Nasal aspirate; PCR	NA	NA	NA	NA	1723	2.4
p1489	Seoul, South Korea (1996-1998) [131]	ALRI	NPA; Culture	NA	NA	250	0.4	NA	NA
p1492	Seoul, South Korea (2006-2007) [132]	ALRI	NPA; PCR	NA	NA	75	4	NA	NA
p1520	multisies, Spain (2011-2013) [133]	ALRI	NP specimen; PCR	NA	NA	40	5	NA	NA
p1557	Istanbul, Turkey (2010-2011) [135]	ARI	NPA; PCR	11	9.1	NA	NA	NA	NA
p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	NA	NA	NA	NA	683	5.1
p1733	Ho Chi Minh, Vietnam (2004-2008) [137]	ARI	NS, TS, and NPA; PCR	NA	NA	13	0	NA	NA
p1870	Cairo, Egypt (2010-2014) [139]	ALRI	NPS and OPS; PCR	NA	NA	NA	NA	191	3.1
p1873	Lanzhou, China (2004-2005) [140]	CXR-pneumonia	NPA; DFA	NA	NA	133	6.8	NA	NA
p1901	Chengdu, China (2007) [141]	ALRI	NP secretion; DFA	NA	NA	121	10.7	NA	NA
p1931	Changsha, China (2015) [142]	ARI	NP secretion; DFA	NA	NA	285	1.4	NA	NA
p216	Beijing and Shandong, China (2012-2015) [143]	ALRI	NPS, NPA and sputum; PCR	NA	NA	850	5.3	NA	NA

* ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

† NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

‡ When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-11 years were re-classified as 5-9 years; 5-15 years, 5-16 years, 6-12 years, 6-15 years, and 6-16 years were classified as 5-14 years; 5-17 years, 5-18 years, 6-17 years, 7-17 years, and 6-18 years were re-classified as 5-19 years.

Study ID	Location (reference)	Case definition	Specimen and test	ALRI cases for 5-9 y	Proportion for 5-9 y	ALRI cases for 5-14 y	Proportion for 5-14 y	ALRI cases for 5-19 y	Proportion for 5-19 y
p221	Beijing, China (2007-2010) [144]	ALRI	NP secretion; PCR	NA	NA	358	3.9	NA	NA
P231	Beijing, China (2014-2016)[190]	ARI AND fever	NPS, TS, or sputum; PCR	NA	NA	157	5.1	NA	NA
p241	Beijing, China (2011-2012) [145]	ALRI	Tracheal aspirate; PCR	NA	NA	NA	NA	93	14
p264	Changsha, China (2007-2008) [146]	ALRI	NPA; PCR	NA	NA	42	4.8	NA	NA
p311	Hebei, China (2007-2008) [147]	ALRI	TS; DFA	NA	NA	361	1.7	NA	NA
p367	Lanzhou,China (2006-2009) [149]	ALRI	NA; PCR	NA	NA	67	6	NA	NA
p376	multisites,China (2009-2013) [150]	ALRI	NS, NA, sputum, BAL or lung puncture; PCR	NA	NA	3510	4.4	NA	NA
p386	Shanghai, China (2003-2006) [151]	ALRI	NPA; DFA	NA	NA	974	1.7	NA	NA
p387	Shanghai,China (2013-2015) [152]	ALRI	NP secretion; DFA	NA	NA	NA	NA	1164	1.1
p396	Shantou, Shenzhen and JIeyang, China (2007-2011) [154]	ARI	NPS or sputum specimen; PCR	NA	NA	112	8	NA	NA
p399	Shantou, Shenzhen and JIeyang, China (2006-2008) [155]	ALRI	PCR; PCR	NA	NA	30	0	NA	NA
p400	nine cities, North China (2015) [156]	ALRI	NP secretion; PCR	NA	NA	331	4.8	NA	NA
p404	Shenzhen, China (2012-2015) [157]	ARI AND fever	NS; DFA	NA	NA	1568	0.3	NA	NA
p421	Tianjin, China (2015-2016) [158]	ARI AND fever	TS; PCR	NA	NA	62	1.6	NA	NA
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	OPS and NPS; PCR	NA	NA	NA	NA	326	4
p52	Perth, Australia (2000-2005) [163]	ALRI	NPA; PCR	97	2.1	NA	NA	NA	NA
p523	Kiel, Germany (1995-1999) [164]	ALRI	NPA; PCR	NA	NA	36	2.8	NA	NA
p540	Athens, Greece (pre 2003) [165]	CXR-pneumonia	NPW; PCR	NA	NA	75	8	NA	NA
p612	Odisha, India (2012-2014) [166]	ARI	multiple; PCR	NA	NA	104	0	NA	NA
p671	Naples, Italy (2016-2017) [169]	ARI	NPS; PCR	36	5.6	NA	NA	NA	NA
p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	NA	149	5.4	NA	NA
p829	Multi-country (2008-2009) [175]	ARI AND fever	multiple; PCR	NA	NA	125	2.4	NA	NA
p953	Lørenskog, Norway (2012-2014) [176]	CXR-pneumonia	NA; NA	NA	NA	NA	NA	40	2.5

Table S5.10 Details of studies with data on hospital admission rates of human metapneumovirus-associated ALRI and human parainfluenza-virus-associated ALRI (per 1,000 children and adolescents per year) [*†‡]

Virus	Study ID	Location (reference)	Case definition	Specimen and test	5-9 y	5-14 y	5-19 y
Human metapneumovirus	p1487	Soweto, South Africa (2009-2013) [191]	ALRI	NA; PCR	NA	NA	0
	p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	0.1	NA	0
	p1631	Salt Lake City Utah, US (2007-2013) [192]	ALRI	NA; DFA and culture	NA	NA	0.1
	p561	Santa Rosa and Quetzaltenango, Guatemala (2007-2012) [193]	ARI	OPS and NPS; PCR	NA	NA	0
	p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	0	NA
	p957	Trondheim, Norway (2006-2015) [181]	ALRI	NPA; PCR	NA	0.1	NA
Human parainfluenza virus	p1537	Sa Kaeo and Nakhon Phanom, Thailand (2003-2007) [194]	ARI	NPS; PCR and serology	0.2	NA	0.1
	p1612	Memphis, Nashville, Salt Lake City, US (2010-2012) [104]	CXR-pneumonia	NPS and OPS and serum; PCR and serology	0.1	NA	0
	p1618	US (1996-1998) [105]	ALRI	mostly NPS; PCR, culture and EIA (enzyme immunoassay)	NA	NA	0.2
	p296	China, Hong Kong SAR (1998-2012) [191]	ARI	NPA; Immunofluorescence	0.4	0.2	NA
	p328	China, Hong Kong SAR (2003-2006) [116]	ARI AND fever	NPA; DFA and culture	0.6	0.3	0.2
	p738	Kilifi, Kenya (2007-2010) [124]	ALRI	NPW or NPA; PCR	NA	0	NA

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-17 years and 5-18 years were re-classified as 5-19 years.

Table S5.12 Details of all studies with data on in-hospital case-fatality ratios (hCFRs) of influenza-associated ALRI. [*†‡]

Study ID	Location (reference)	Case definition	Cases for 5-9 y	hCFR for 5-9 y (%)	Cases for 5-14 y	hCFR for 5-14 y (%)	Cases for 5-19 y	hCFR for 5-19 y (%)
p1070	Singapore (2013-2014) [97]	ARI AND fever	270	0.7	NA	NA	418	0.7
p146	multisites, Canada (2004-2005) [195]	ARI	NA	NA	80	1.2	NA	NA
p1512	Catalonia, Spain (2010-2015) [196]	ALRI	NA	NA	46	2.2	NA	NA
p1518	Spain (2009-2015) [197]	ICD-coded lab-confirmed influenza	NA	NA	1525	1	NA	NA
p1519	Spain (2010-2016) [99]	ALRI	NA	NA	644	3.1	NA	NA
p1565	multisites, Uganda (2010-2015) [86]	ARI	NA	NA	20	5	NA	NA
p1637	Salt Lake County, Utah, US (2001-2004) [107]	ARI	NA	NA	NA	NA	34	2.9
p212	multisites, Canada (2003-2004) [198]	ARI	NA	NA	NA	NA	82	1.2
P23	Australia (2014)[199]	ARI	0	0	1	2.0	NA	NA
P244	China (2010-2011)[200]	ARI AND fever	NA	NA	109	2.8	NA	NA
p296	China, Hong Kong SAR (1998-2012) [191]	ARI	2384	0	3075	0.1	NA	NA
p466	multisites, Egypt (2007-2014) [90]	ARI AND fever	132	0.8	195	1	264	2.3
p491	Turku, Finland (1988-2004) [119]	ARI	NA	NA	105	1	NA	NA
p520	Germany (2004-2010) [120]	ICD-coded lab-confirmed influenza	NA	NA	8189	0.5	NA	NA
p58	Sydney and Perth, Australia (2011-2013) [201]	ARI	185	1.1	264	0.8	NA	NA
p611	Madhya Pradesh, India (2009-2015) [202]	ARI AND fever	NA	NA	116	1.7	NA	NA
p624	East Jakarta, Indonesia (2011-2014) [168]	ARI AND fever	NA	NA	NA	NA	111	0
p655	Ireland (2010-2011) [123]	ARI AND fever	53	3.8	91	3.3	NA	NA
p679	multisites, Jordan (2008-2014) [170]	ALRI	NA	NA	31	3.2	NA	NA
p749	Mexico (2013-2014) [203]	ALRI	NA	NA	127	7.1	NA	NA
p758	Mexico city, Mexico (2011-2012) [204]	ALRI	NA	NA	95	3.2	NA	NA
p794	eight African countries (2009-2012) [93]	ARI AND fever	NA	NA	NA	NA	320	0
p804	Nine eastern Europe countries (2009-2012) [205]	ALRI	NA	NA	207	1.9	NA	NA

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

[‡] When data were reported for other age bands, data were grouped into the closest one of the above age bands. For example, 5-11 years were re-classified as 5-9 years; 5-15 years and 6-14 years were classified as 5-14 years; 5-17 years, and 5-18 years were re-classified as 5-19 years.

Study ID	Location (reference)	Case definition	Cases for 5-9 y	hCFR for 5-9 y (%)	Cases for 5-14 y	hCFR for 5-14 y (%)	Cases for 5-19 y	hCFR for 5-19 y (%)
p818	Four countries (Russia, Turkey, China, Spain) (2013-2014) [206]	ARI AND one systemic symptom	NA	NA	NA	NA	102	0
p963	Sohar, Oman (2010-2013) [125]	ARI	NA	NA	53	1.9	NA	NA
p983	Oman (2012-2015) [74]	ARI	NA	NA	519	1.3	NA	NA

Table S5.13 Estimates of in-hospital case-fatality ratios (hCFRs) of influenza-associated ALRI and in-hospital mortality in children and adolescents aged 5-14 years, by World Bank income region.

	World Bank income region	No. of studies	hCFR (%)[§]	In-hospital mortality for 5-14 years[**]
Using hCFR data for 5-14 years or 5-19 years[††]	Low & lower middle	5	0.8 (0.2-3.3)	3700 (500-27100)
	Upper middle	6	2.6 (1.3-5.2)	8200 (2300-27600)
	High	15	0.9 (0.5-1.6)	200 (100-500)
	Global			12100 (2900-55200)
Using hCFR data for 5-14 years[‡‡]	Low & lower middle	3	1.5 (0.6-3.6)	7100 (1600-29600)
	Upper middle	5	3.4 (2-5.8)	10700 (3500-31600)
	High	12	0.9 (0.5-1.7)	200 (100-500)
	Global			18000 (5200-61700)

Given the paucity of influenza-specific hCFR data, we did a primary analysis for influenza-associated ALRI in-hospital deaths for 5-14 years. Table S6.11 shows the preliminary estimates. An estimated 10,500 (UR 2,200-49,800) influenza-associated ALRI in-hospital deaths occurred among children and adolescents aged 5-14 years globally when including all hCFR data (for 5-14 years or 5-19 years). When restricted the analysis to including only hCFR data for 5-14 years, we estimated 18,900 (UR 4,700-75,600) influenza-associated ALRI in-hospital deaths globally for 5-14 years. We were unable to estimate influenza-associated ALRI in-hospital deaths for 5-19 years due to the lack of data. We were unable to estimate in-hospital deaths for the other three viruses given the paucity of data on hCFRs of respiratory syncytial virus (four studies), human metapneumovirus (one study), and human parainfluenza virus (one study).

[§] Estimates from meta-analyses.
[**] Estimated by combining hCFRs and hospital admissions of IFV-associated ALRI by regions.
[††] hCFR estimates were based on all data for 5-14 years or 5-19 years.
[‡‡] hCFR estimates were based on the data for 5-14 years.

Table S5.14 Details of all studies with data on in-hospital case-fatality ratios (hCFRs) of ALRI associated with respiratory syncytial virus, human metapneumovirus, and human parainfluenza virus. [*†]

	Study ID	Location (reference)	Case definition	Age	Virus-confirmed cases	hCFR (%)
Respiratory syncytial virus	p296	China, Hong Kong SAR (1998-2012) [191]	ARI	5-14 y	318	2.2
	p562	Santa Rosa and Quetzaltenango, Guatemala (2008-2012)[180]	ARI	5-17 y	30	6.67
	p752	Mexico city, Mexico (2004-2008)[207]	ARI	5-18 y	20	10
	p1511	Seremban, Malaysia (2008-2013)[208]	ALRI	5-16 y	12	0
Human metapneumovirus	p1611	Columbus, Ohio, US (2007-2010)[209]	ARI	5-15 y	23	0
Human parainfluenza virus	p296	China, Hong Kong SAR (1998-2012) [191]	ARI	5-14 y	3075	0
	p1567	England and Wales, UK (1989-1997)[210]	ALRI	5-14 y	241	2.9

[*] ARI: any respiratory symptoms (e.g., cough, runny nose, sore throat, shortness of breath, or difficulty in breathing) requiring hospital admission. ARI AND fever: fever or history of fever AND any respiratory symptoms requiring hospital admission. ALRI: any of physician-diagnosed pneumonia (and bronchiolitis), ICD-coded pneumonia (and bronchiolitis), or fever or history of fever AND cough or sour throat AND shortness of breath or difficulty breathing. CXR-pneumonia: chest radiograph confirmed pneumonia. y: years.

[†] NP: nasopharyngeal. NPS: nasopharyngeal swab. OPS: oropharyngeal swab. PCR: polymerase chain reaction. NS: nasal swab. DFA: direct fluorescent antibody test. EIA: enzyme immunoassay. SPIA: solid-phase immunoassay. TS: throat swab. NPA: nasopharyngeal aspirate. NPW: nasopharyngeal wash. BAL: bronchoalveolar lavage.

Appendix 6 Assessment of risk of bias

Table S6.1 Assessment criteria

Category	Description	Risk of bias
Study design	Studies where the cases were prospectively enrolled	Low
	Other study designs	High
Adjustment for healthcare utilization (only for hospital admission rate studies)	Meeting either of the following: 1. Including all or main hospitals in the catchment area; 2. Not including main hospitals, but adjusting for the proportion of patients admitted in the study hospitals	Low
	Not including main hospitals OR no related description; AND no adjustment for the proportion of patients admitted in the study hospitals	High
Patient groups excluded	No exclusions that may affect estimates	Low
	Exclusions that may affect estimates, such as: 1. Excluding children and adolescents with high-risk conditions. 2. Other exclusions that may affect estimates	High
Case definition	Any of the following: 1. physician-diagnosed ALRI or pneumonia 2. ALRI diagnosed based on clinical symptoms or signs, e.g., fever AND cough or sore throat AND shortness of breath or difficulty breathing 3. chest radiograph confirmed pneumonia 4. ARI requiring hospital admission; ARI AND fever requiring hospital admission.	Low
	Other definitions that are more specific or less specific, e.g., ARI or fever requiring hospital admission; hospitalised with infection infections (ICD-code)	High
Sampling strategy (only for studies with data on viruses)	The proportion of cases tested is available AND either of the following: 1. >=90% of eligible patients are tested. 2. Testing a systematic/random sample of patients.	Low
	<90% of eligible cases are tested; OR The proportion of cases that are tested is unavailable.	High
Test (only for studies with data on viruses)	PCR; Or using other diagnostic tests, but confirming negative samples with PCR	Low
	1. Other diagnostic tests, e.g., culture, IFA, DFA. 2. No mention of test methods	High

Table S6.2 Risk of bias for studies reporting data on ALRI hospital admission rates.

Study ID	Location; period	Study design	Adjustment for healthcare utilisation	Patient groups excluded	Case definition
p45	Mornington, Australia; 2007-2011[43]	High	High	Low	Low
p55	Western Australia; 1996-2012[48]	High	Low	Low	Low
p62	Brazil; 2003-2004[53]	High	Low	Low	Low
p73	Brazil; 2005-2015 [64]	High	Low	Low	Low
p94	La Paz, Bolivia; 2012-2017[71]	Low	Low	Low	Low
p99	Svay Rieng, Cambodia; 2015-2016 [75]	Low	Low	Low	Low
p99	Siem Reap, Cambodia; 2015-2016 [75]	Low	Low	Low	Low
p99	Kampong Cham, Cambodia; 2015-2016 [75]	Low	Low	Low	Low
p113	Canada; 2004-2010[10]	High	Low	Low	Low
p117	Edmonton, Canada; 2000-2002[11]	Low	High	High	Low
p125	multisites, Canada; 1995-2001[12]	High	Low	Low	Low
p165	Ontario, Canada; 1992-2014[31]	High	Low	Low	Low
p226	Beijing, China; 2017-2018[34]	Low	Low	Low	Low
p286	China, Hong Kong SAR; 2005[35]	High	Low	Low	Low
p321	China, Hong Kong SAR; 2011-2015[36]	High	Low	Low	Low
p355	Jingzhou, China; 2010-2012[37]	Low	Low	Low	Low
p416	Suzhou, China; 2010-2014[38]	High	Low	Low	Low
p420	Taiwan, China; 1997-2004[39]	High	Low	Low	Low
p433	Denmark; 1995-1999[40]	High	Low	Low	Low
p440	Denmark; 1997-2011[41]	High	Low	Low	Low
p447	Ecuador; 2011-2015 [42]	High	Low	Low	Low
p454	Damanhour, Egypt; 2009-2012[44]	Low	Low	Low	Low
p509	Rhône-Alpes, France; 2005-2010[45]	High	Low	Low	Low
p524	Kiel, Germany; 1996-1999[46]	Low	Low	Low	Low
p532	Shai-Osudoku and Ningo-Prampram, Ghana; 2013-2015[47]	Low	Low	Low	Low
p557	Santa Rosa and Quetzaltenango, Guatemala; 2007-2011[49]	Low	Low	Low	Low
p566	Hungary; 2006-2011[50]	High	Low	Low	Low
p567	Ballabgarh, India; 2010-2012[51]	Low	Low	Low	Low
p567	Vadu, India; 2010-2012[51]	Low	Low	Low	Low
p572	Ballabgarh, India; 2012-2014[52]	Low	Low	Low	Low
p630	Wonosari, Indonesia; 2013-2016[54]	Low	Low	Low	Low
p630	Kanudjoso Djati, Indonesia; 2013-2016[54]	Low	Low	Low	Low
p630	Deli Serdang, Indonesia; 2013-2016[54]	Low	Low	Low	Low
p633	Iran; 2015 [55]	High	High	Low	Low
p646	multisites, Iran; 2012-2013 [56]	Low	Low	Low	Low
p669	Negev, Israel; 1989-1991[57]	High	Low	Low	Low
p672	Veneto, Italy; 2004-2012[58]	High	Low	Low	Low
p689	Bondo, Kenya; 2001-2003[59]	High	Low	Low	Low

Study ID	Location; period	Study design	Adjustment for healthcare utilisation	Patient groups excluded	Case definition
p693	Bondo, Kenya; 2007-2009[60]	Low	Low	Low	Low
p704	Dadaab, Kenya; 2010-2012[61]	Low	Low	Low	Low
p705	Karemo, Kenya; 2009-2012[62]	Low	Low	Low	Low
p706	Kenya; 2012-2014[63]	low	Low	Low	Low
p731	Kilifi, Kenya; 2002-2015[65]	Low	Low	Low	Low
p748	Blantyre, Malawi; 2011-2014[66]	Low	Low	Low	Low
p774	multisites, Mongolia; 2007-2012[67]	Low	Low	Low	Low
p877	Netherlands; 2001-2007[68]	High	Low	Low	Low
p882	Netherlands; 2008-2011[69]	High	Low	Low	Low
p939	Auckland, New Zealand; 1993-1996[70]	Low	Low	Low	Low
p945	Leo´n, Nicaragua; 2008-2015[72]	Low	Low	Low	Low
p954	Norway; 2008-2009[73]	High	Low	Low	Low
p983	Oman; 2012-2015[74]	High	Low	Low	Low
p1000	Baguio, Philippines; 2010-2011[7]	Low	Low	Low	Low
p1006	Baguio, Philippines; 2012-2014[8]	Low	Low	Low	Low
p1015	Chrzanów County, Poland; 2006-2008[9]	High	Low	Low	Low
p1015	Inowroclaw County, Poland; 2006-2008[9]	High	Low	Low	Low
p1483	South Africa; 2007-2012[13]	High	Low	Low	Low
p1503	South Korea; 2002-2005[14]	Low	Low	Low	Low
p1505	South Korea; 2007-2014[15]	High	Low	Low	Low
p1516	Spain; 1995-1998[16]	High	Low	Low	Low
p1517	Spain; 2001-2014[17]	High	Low	Low	Low
p1522	Sweden; 2005[18]	High	Low	Low	Low
p1532	Sa Kaeo, Thailand; 2002-2003[19]	Low	Low	Low	Low
p1541	Sa Kaeo and Nakhon Phanom, Thailand; 2003-2007[20]	Low	Low	Low	Low
p1546	Sa Kaeo and Nakhon Phanom, Thailand; 2008-2011[21]	Low	Low	Low	Low
p1548	Sfax, Tunisia; 2003-2015[22]	High	High	Low	Low
p1566	Wakiso, Uganda; 2013-2016[23]	Low	Low	Low	Low
p1566	Mbarara, Uganda; 2013-2016[23]	Low	Low	Low	Low
p1566	Tororo, Uganda; 2013-2016[23]	Low	Low	Low	Low
p1571	England, UK; 2001-2013[24]	High	Low	Low	Low
p1577	North East England, UK; 2001-2002[25]	Low	Low	Low	Low
p1578	North East England, UK; 2008-2009[26]	Low	Low	Low	Low
p1583	Scotland, UK; 1981-2005[27]	High	Low	Low	Low
p1587	Scotland, UK; 2000-2012[28]	High	Low	Low	Low
p1604	two municipalities, Uruguay; 2009-2012[29]	Low	High	Low	Low
p1640	US; 1996-2011[30]	High	Low	Low	Low
p1654	US; 2012-2014[32]	High	Low	Low	Low
p1871	Beijing, China; 2014-2016[33]	Low	Low	Low	Low

Table S6.3 Risk of bias for studies with data on influenza-associated ALRI hospital admission rates

Study ID	Location; period	Study design	Adjustment for healthcare utilisation	Patient groups excluded	Case definition	Sampling strategy	Test
p39	Australia; 2006-2015[118]	High	Low	Low	High	High	High
p94	La Paz, Bolivia; 2012-2017[71]	Low	Low	Low	Low	Low	Low
p99	Svay Rieng, Cambodia; 2015-2016 [75]	Low	Low	Low	Low	High	Low
p99	Siem Reap, Cambodia; 2015-2016 [75]	Low	Low	Low	Low	High	Low
p99	Kampong Cham, Cambodia; 2015-2016 [75]	Low	Low	Low	Low	High	Low
p105	Canada; 2003-2008[96]	High	Low	Low	High	High	High
p108	Canada; 2003-2007; 2010-2014[98]	High	Low	Low	Low	High	High
p226	Beijing, China; 2017-2018[34]	Low	Low	Low	Low	High	Low
p328	China, Hong Kong SAR; 2003-2006[116]	Low	Low	High	Low	Low	High
p330	China, Hong Kong SAR; 2005-2011[117]	High	Low	Low	High	High	High
p355	Jingzhou, China; 2010-2012[37]	Low	Low	Low	Low	Low	High
p491	Turku, Finland; 1988-2004[119]	High	Low	Low	Low	High	High
p520	Germany; 2004-2010[120]	High	Low	Low	High	High	High
p525	Kiel, Germany; 1996-2001[121]	Low	Low	Low	Low	High	Low
p532	Shai-Osudoku and Ningo-Prampram, Ghana; 2013-2015[47]	Low	Low	Low	Low	High	Low
p536	Athens, Greece; 2002-2005[122]	Low	Low	Low	High	High	Low
p567	Ballabgarh, India; 2010-2012[51]	Low	Low	Low	Low	Low	Low
p567	Vadu, India; 2010-2012[51]	Low	Low	Low	Low	Low	Low
p630	Wonosari, Indonesia; 2013-2016[54]	Low	Low	Low	Low	Low	Low
p630	Kanudjoso Djati, Indonesia; 2013-2016[54]	Low	Low	Low	Low	Low	Low
p630	Deli Serdang, Indonesia; 2013-2016[54]	Low	Low	Low	Low	Low	Low
p655	Ireland; 2010-2011[123]	High	Low	Low	Low	High	Low
p693	Bondo, Kenya; 2007-2009[60]	Low	Low	Low	Low	High	Low
p704	Dadaab, Kenya; 2010-2012[61]	Low	Low	Low	Low	Low	Low
p705	Karemo, Kenya; 2009-2012[62]	Low	Low	Low	Low	High	Low
p738	Kilifi, Kenya; 2007-2010[124]	Low	Low	Low	Low	High	Low
p748	Blantyre, Malawi; 2011-2014[66]	Low	Low	Low	Low	Low	Low
p963	Sohar, Oman; 2010-2013[125]	Low	Low	Low	Low	Low	Low

Study ID	Location; period	Study design	Adjustment for healthcare utilisation	Patient groups excluded	Case definition	Sampling strategy	Test
p963	Ibra, Oman; 2010-2013[125]	Low	Low	Low	Low	Low	Low
p963	SQH, Oman; 2010-2013[125]	Low	Low	Low	Low	Low	Low
p983	Oman; 2012-2015[74]	High	Low	Low	Low	High	Low
p995	multisites, Peru; 2009-2015[126]	Low	Low	Low	Low	Low	Low
p1000	Baguio, Philippines; 2010-2011[7]	Low	Low	Low	Low	Low	Low
p1006	Baguio, Philippines; 2012-2014[8]	Low	Low	Low	Low	Low	Low
p1070	Singapore; 2013-2014[97]	High	Low	Low	Low	High	High
p1519	Spain; 2010-2016[99]	High	Low	Low	Low	High	High
p1525	Basel, Switzerland; 2001-2002[100]	High	High	High	Low	High	High
p1545	Sa Kaeo and Nakhon Phanom, Thailand; 2005-2008 Incidence, Seasonality and Mortality Associated with Influenza Pneumonia in Thailand: 2005-2008.	Low	Low	Low	Low	High	Low
p1566	Wakiso, Uganda; 2013-2016[23]	Low	Low	Low	Low	High	Low
p1566	Mbarara, Uganda; 2013-2016[23]	Low	Low	Low	Low	High	Low
p1566	Tororo, Uganda; 2013-2016[23]	Low	Low	Low	Low	High	Low
p1574	England, UK; 2010-2015 [102]	High	Low	Low	Low	High	High
p1610	Colorado, US; 2004-2008[103]	High	Low	Low	Low	High	High
p1612	Memphis, Nashville, Salt Lake City, US; 2010-2012[104]	Low	Low	High	Low	Low	High
p1618	US; 1996-1998[105]	Low	Low	Low	Low	High	High
p1625	Philadelphia, US; 2000-2004[106]	High	Low	Low	Low	High	High
p1637	Salt Lake County, Utah, US; 2001-2004[107]	High	Low	Low	Low	High	High
p1656	US; 2003-2008[108]	Low	Low	Low	Low	High	Low
p1668	US; 2008-2011[109]	Low	Low	Low	Low	High	Low
p1671	US; 2012-2013[110]	Low	Low	Low	Low	High	Low
p1672	US; 2013-2014[111]	Low	Low	Low	Low	High	Low
p1686	US; 2017-2018[112]	Low	Low	Low	Low	High	Low
p1690	US; 2018-2019[113]	Low	Low	Low	Low	High	Low
p1700	Utah, US; 2016-2017[114]	Low	Low	Low	Low	High	Low
p1735	Nha Trang, Viet Nam; 2007-2012[115]	Low	Low	Low	Low	High	Low

Study ID	Location; period	Study design	Adjustment for healthcare utilisation	Patient groups excluded	Case definition	Sampling strategy	Test
p1871	Beijing, China; 2014-2016[33]	Low	Low	Low	Low	Low	Low
p774	multisites, Mongolia; 2007-2012[67]	Low	Low	Low	Low	Low	High

Table S6.4 Risk of bias for studies with data on in-hospital case-fatality ratios (hCFRs) of ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition
p1015	Chrzanów County, Poland; 2006-2008[9]	High	Low	Low
p1071	multisites, South Africa; 2009-2012[83]	Low	Low	Low
p113	Canada; 2004-2010[10]	High	Low	Low
p1505	South Korea; 2007-2014[15]	High	Low	Low
p1513	Spain; 1995-1996[84]	High	Low	Low
p1527	Sa Kaeo; 2002-2003[85]	Low	Low	Low
p1541	Sa Kaeo and Nakhon Phanom, Thailand; 2003-2007[20]	Low	Low	Low
p1548	Sfax, Tunisia; 2003-2015[22]	High	Low	Low
p1565	multisites, Uganda; 2010-2015[86]	Low	Low	Low
p1568	England, UK; 2000-2007[87]	High	Low	Low
p1602	Paysandú and Salto, Uruguay; 2001-2004[88]	Low	Low	Low
p1654	US; 2012-2014[32]	High	Low	Low
p246	Guangzhou, China; 2009-2012[89]	High	Low	Low
p321	China, Hong Kong SAR; 2011-2015[36]	High	Low	Low
p420	Taiwan, China; 1997-2004[39]	High	Low	Low
p440	Denmark; 1997-2011[41]	High	Low	Low
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low
p566	Hungary; 2006-2011[50]	High	Low	Low
p62	Brazil; 2003-2007[53]	High	Low	Low
p633	Iran; 2015[55]	High	Low	Low
p689	Bondo, Kenya; 2001-2003[59]	High	Low	Low
p739	multisites, Kenya; 2014-2018[91]	High	Low	Low
p780	Maputo, Mozambique; 2014-2016[92]	Low	Low	Low
p794	eight African countries; 2009-2012[93]	Low	Low	Low
p90	Rio de Janeiro, Brazil; 1996-2011[94]	Low	Low	Low
p983	Oman; 2012-2015[74]	High	Low	Low
p992	Asuncion, Paraguay; 2004-2013[95]	High	Low	Low

Table S6.5 Risk of bias for studies with data on proportion positives of influenza in hospitalised ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p52	Perth, Australia; 2000-2005[163]	High	Low	Low	High	Low
p101	Yaounde, Cameroon; 2011-2013[128]	Low	Low	Low	Low	Low
p216	Beijing and Shandong, China; 2012-2015[143]	Low	Low	Low	High	Low
p221	Beijing, China; 2007-2010[144]	Low	Low	Low	High	Low
p241	Beijing, China; 2011-2012[145]	High	Low	Low	High	Low
p264	Changsha, China; 2007-2008[146]	Low	Low	Low	High	Low
p311	Hebei, China; 2007-2008[147]	Low	Low	Low	Low	High
p331	China, Hong Kong SAR; 2009-2013[148]	Low	Low	Low	Low	High
p367	Lanzhou,China; 2006-2009[149]	High	Low	Low	Low	Low
p376	multisites,China; 2009-2013[150]	Low	Low	Low	Low	Low
p386	Shanghai, China; 2003-2006[151]	High	Low	Low	High	High
p387	Shanghai,China; 2013-2015[152]	Low	Low	Low	Low	High
p388	Shanghai, China; 2016-2017[153]	Low	Low	Low	Low	Low
p399	Shantou, Shenzhen and Jieyang, China; 2006-2008[155]	Low	Low	Low	High	Low
p400	nine cities, North China; 2015[156]	Low	Low	Low	High	Low
p404	Shenzhen, China; 2012-2015[157]	Low	Low	Low	High	High
p421	Tianjin, China; 2015-2016[158]	Low	Low	Low	High	Low
p427	Zhejiang, China; 2011-2015[159]	Low	Low	Low	High	Low
p428	multisites, DR Congo; 2015[160]	Low	Low	Low	Low	Low
p452	Damanhour, Egypt; 2013[161]	Low	Low	Low	Low	Low
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low	Low	Low
p476	multisites, Egypt; 2012-2015[162]	Low	Low	Low	High	Low
p523	Kiel, Germany; 1995-1999[164]	Low	Low	Low	High	Low
p540	Athens, Greece; (pre 2003) [165]	Low	High	Low	Low	Low
p612	Odisha, India; 2012-2014[166]	Low	Low	Low	High	Low
p619	East Jakarta, Indonesia; 2011-2012[167]	Low	Low	Low	Low	Low
p624	East Jakarta, Indonesia; 2011-2014[168]	Low	Low	Low	Low	Low
p671	Naples, Italy; 2016-2017[169]	High	Low	Low	High	Low
p679	multisites, Jordan; 2008-2014[170]	Low	Low	Low	Low	Low
p744	Southern province, Lebanon; 2015-2016[171]	Low	Low	Low	Low	Low
p744	Beirut, Lebanon; 2015-2016 [171]	Low	Low	Low	Low	Low
p775	Morocco; 2007-2009[172]	Low	Low	Low	High	High
p780	Maputo, Mozambique; 2014-2016[92]	Low	Low	Low	Low	Low
p781	11 countries AFR; 2010-2012[173]	Low	Low	Low	High	Low
p784	14 countries AFR and EMR; 2006-2010 [174]	Low	Low	Low	High	Low
p829	Multi-country; 2008-2009[175]	High	Low	Low	High	Low
p953	Lørenskog, Norway; 2012-2014[176]	Low	High	Low	Low	Low
p1018	Riyadh, Saudi Arabia; 2005-2010[129]	High	Low	Low	High	High
p1071	multisites, South Africa; 2009-2012[83]	Low	Low	Low	Low	Low
p1488	Seongnam, South Korea; 2006-2016[130]	High	Low	Low	High	Low

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p1489	Seoul, South Korea; 1996-1998[131]	Low	Low	Low	High	High
p1492	Seoul, South Korea; 2006-2007[132]	Low	Low	Low	High	Low
p1520	multisies, Spain; 2011-2013[133]	Low	High	Low	Low	Low
p1520	London, UK; 2011-2013 [133]	Low	High	Low	Low	Low
p1554	Bursa, Turkey; 2015-2018[134]	High	Low	Low	High	Low
p1557	Istanbul, Turkey; 2010-2011[135]	High	Low	Low	High	Low
p1565	multisites, Uganda; 2010-2015[86]	Low	Low	Low	High	Low
p1733	Ho Chi Minh, Vietnam; 2004-2008[137]	Low	Low	Low	High	Low
p1739	Vietnam; 2011-2014[138]	Low	Low	Low	High	Low
p1870	Cairo, Egypt; 2010-2014[139]	Low	Low	Low	High	Low
p1873	Lanzhou, China; 2004-2005[140]	Low	Low	Low	Low	High
p1901	Chengdu, China; 2007[141]	Low	Low	Low	Low	High
p1931	Changsha, China; 2015[142]	Low	Low	Low	Low	High
p396	Shantou, Shenzhen and JIeyang, China; 2007-2011[154]	Low	Low	Low	High	Low

Table S6.5 Risk of bias for studies with data on in-hospital case-fatality ratios (hCFRs) of influenza-associated ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p1070	Singapore; 2013-2014[97]	High	Low	Low	High	High
p146	multisites, Canada; 2004-2005[195]	High	Low	Low	High	High
p1512	Catalonia, Spain; 2010-2015[196]	High	Low	Low	High	High
p1518	Spain; 2009-2015[197]	High	Low	High	High	High
p1519	Spain; 2010-2016[99]	High	Low	Low	High	High
p1565	multisites, Uganda; 2010-2015[86]	Low	Low	Low	High	Low
p1637	Salt Lake County, Utah, US; 2001-2004[107]	High	Low	Low	High	High
p212	multisites, Canada; 2003-2004[198]	High	Low	Low	High	High
p296	China, Hong Kong SAR; 1998-2012[191]	High	Low	Low	High	High
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low	Low	Low
p491	Turku, Finland; 1988-2004[119]	High	Low	Low	High	High
p520	Germany; 2004-2010[120]	High	Low	High	High	High
p58	Sydney and Perth, Australia; 2011-2013[201]	High	Low	Low	High	Low
p611	Madhya Pradesh, India; 2009-2015[202]	High	Low	Low	High	Low
p624	East Jakarta, Indonesia; 2011-2014[168]	Low	Low	Low	Low	Low
p655	Ireland; 2010-2011[123]	High	Low	Low	High	Low
p679	multisites, Jordan; 2008-2014[170]	Low	Low	Low	Low	Low
p749	Mexico; 2013-2014[203]	High	Low	Low	High	Low
p758	Mexico city, Mexico; 2011-2012[204]	High	Low	Low	High	Low
p794	eight African countries; 2009-2012[93]	Low	Low	Low	High	Low
p804	Nine eastern Europe countries; 2009-2012[205]	Low	Low	Low	Low	High
p818	Four countries (Russia, Turkey, China, Spain); 2013-2014[206]	Low	Low	Low	Low	Low
p963	Sohar, Oman; 2010-2013[125]	Low	Low	Low	Low	Low
p983	Oman; 2012-2015[74]	High	Low	Low	High	Low

Table S6.5 Risk of bias for studies with data on proportion positives of respiratory syncytial virus in hospitalised ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p52	Perth, Australia; 2000-2005[163]	High	Low	Low	High	Low
p94	La Paz, Bolivia; 2012-2017[71]	Low	Low	Low	Low	Low
p101	Yaounde, Cameroon; 2011-2013[128]	Low	Low	Low	Low	Low
p216	Beijing and Shandong, China; 2012-2015[143]	Low	Low	Low	High	Low
p221	Beijing, China; 2007-2010[144]	Low	Low	Low	High	Low
p241	Beijing, China; 2011-2012[145]	High	Low	Low	High	Low
p245	Chongqing, China; 2006-2008[184]	Low	Low	Low	Low	Low
p264	Changsha, China; 2007-2008[146]	Low	Low	Low	High	Low
p311	Hebei, China; 2007-2008[147]	Low	Low	Low	Low	High
p367	Lanzhou,China; 2006-2009[149]	High	Low	Low	Low	Low
p376	multisites,China; 2009-2013[150]	Low	Low	Low	Low	Low
p386	Shanghai, China; 2003-2006[151]	High	Low	Low	High	High
p387	Shanghai,China; 2013-2015[152]	Low	Low	Low	Low	High
p388	Shanghai, China; 2016-2017[153]	Low	Low	Low	Low	Low
p396	Shantou, Shenzhen and Jieyang, China; 2007-2011[154]	Low	Low	Low	High	Low
p399	Shantou, Shenzhen and Jieyang, China; 2006-2008[155]	Low	Low	Low	High	Low
p400	nine cities, North China; 2015[156]	Low	Low	Low	High	Low
p404	Shenzhen, China; 2012-2015[157]	Low	Low	Low	High	High
p421	Tianjin, China; 2015-2016[158]	Low	Low	Low	High	Low
p454	Damanhour, Egypt; 2009-2012[44]	Low	Low	Low	Low	Low
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low	Low	Low
p476	multisites, Egypt; 2012-2015[162]	Low	Low	Low	High	Low
p523	Kiel, Germany; 1995-1999[164]	Low	Low	Low	High	Low
p524	Germany; 1996-1999[46]	Low	Low	Low	High	Low
p540	Athens, Greece; (pre 2003) [165]	Low	High	Low	Low	Low
p562	Guatemala; 2008-2012[180]	Low	Low	Low	High	Low
p587	Lucknow, India; 2011-2013[185]	Low	Low	Low	High	Low
p612	Odisha, India; 2012-2014[166]	Low	Low	Low	High	Low
p671	Naples, Italy; 2016-2017[169]	High	Low	Low	High	Low
p705	Karemo, Kenya; 2009-2012[62]	Low	Low	Low	High	Low
p738	Kilifi, Kenya; 2007-2010[124]	Low	Low	Low	High	Low
p742	Nairobi, Kenya, Slum; 2007-2011[186]	Low	Low	Low	High	Low
p829	Multi-country; 2008-2009[175]	High	Low	Low	High	Low
p953	Lørenskog, Norway; 2012-2014[176]	Low	High	Low	Low	Low
p1000	Baguio, Philippines; 2010-2011[7]	Low	Low	Low	Low	Low
p1006	Baguio, Philippines; 2012-2014[8]	Low	Low	Low	Low	Low
p1018	Riyadh, Saudi Arabia; 2005-2010[129]	High	Low	Low	High	High
p1071	multisites, South Africa; 2009-2012[83]	Low	Low	Low	Low	Low
p1488	Seongnam, South Korea; 2006-2016[130]	High	Low	Low	High	Low

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p1489	Seoul, South Korea; 1996-1998[131]	Low	Low	Low	High	High
p1492	Seoul, South Korea; 2006-2007[132]	Low	Low	Low	High	Low
p1520	multisies, Spain; 2011-2013[133]	Low	High	Low	Low	Low
p1520	London, UK; 2011-2013[133]	Low	High	Low	Low	Low
p1541	Sa Kaeo and Nakhon Phanom, Thailand; 2003-2007[20]	Low	Low	Low	High	Low
p1546	Sa Kaeo and Nakhon Phanom, Thailand; 2008-2011[21]	Low	Low	Low	High	Low
p1554	Bursa, Turkey; 2015-2018[134]	High	Low	Low	High	Low
p1557	Istanbul, Turkey; 2010-2011[135]	High	Low	Low	High	Low
p1612	Memphis, Nashville, Salt Lake City, US; 2010-2012[104]	Low	High	Low	Low	High
p1733	Ho Chi Minh, Vietnam; 2004-2008[137]	Low	Low	Low	High	Low
p1870	Cairo, Egypt; 2010-2014[139]	Low	Low	Low	High	Low
P1872	San Luis Potosı, Mexico; 2009-2010[182]	Low	Low	Low	Low	Low
p1873	Lanzhou, China; 2004-2005[140]	Low	Low	Low	Low	High
p1901	Chengdu, China; 2007[141]	Low	Low	Low	Low	High
p1931	Changsha, China; 2015[142]	Low	Low	Low	Low	High
P1939	Jieyang, China; 2016-2017[183]	Low	Low	Low	High	High

Table S6.6 Risk of bias for studies with data on proportion positives of human metapneumovirus in hospitalised ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p101	Yaounde, Cameroon; 2011-2013[128]	Low	Low	Low	Low	Low
p216	Beijing and Shandong, China; 2012-2015[143]	Low	Low	Low	High	Low
p221	Beijing, China; 2007-2010[144]	Low	Low	Low	High	Low
p241	Beijing, China; 2011-2012[145]	High	Low	Low	High	Low
p245	Chongqing, China; 2006-2008[184]	Low	Low	Low	Low	Low
p257	Guangzhou, China; 2013-2016[187]	Low	Low	Low	High	Low
p264	Changsha, China; 2007-2008[146]	Low	Low	Low	High	Low
p367	Lanzhou,China; 2006-2009[149]	High	Low	Low	Low	Low
p376	multisites,China; 2009-2013[150]	Low	Low	Low	Low	Low
p387	Shanghai,China; 2013-2015[152]	Low	Low	Low	Low	High
p388	Shanghai, China; 2016-2017[153]	Low	Low	Low	Low	Low
p396	Shantou, Shenzhen and JIeyang, China; 2007-2011[154]	Low	Low	Low	High	Low
p399	Shantou, Shenzhen and JIeyang, China; 2006-2008[155]	Low	Low	Low	High	Low
p400	nine cities, North China; 2015[156]	Low	Low	Low	High	Low
p407	Suzhou, China; 2005-2006[188]	High	Low	Low	High	Low
p411	Suzhou, China; 2006-2009[189]	Low	Low	Low	Low	Low
p421	Tianjin, China; 2015-2016[158]	Low	Low	Low	High	Low
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low	Low	Low
p540	Athens, Greece; (pre 2003) [165]	Low	High	Low	Low	Low
p587	Lucknow, India; 2011-2013[185]	Low	Low	Low	High	Low
p612	Odisha, India; 2012-2014[166]	Low	Low	Low	High	Low
p671	Naples, Italy; 2016-2017[169]	High	Low	Low	High	Low
p738	Kilifi, Kenya; 2007-2010[124]	Low	Low	Low	High	Low
p829	Multi-country; 2008-2009[175]	High	Low	Low	High	Low
p953	Lørenskog, Norway; 2012-2014[176]	Low	High	Low	Low	Low
p1071	multisites, South Africa; 2009-2012[83]	Low	Low	Low	Low	Low
p1488	Seongnam, South Korea; 2006-2016[130]	High	Low	Low	High	Low
p1492	Seoul, South Korea; 2006-2007[132]	Low	Low	Low	High	Low
p1520	multisies, Spain; 2011-2013[133]	Low	High	Low	Low	Low
p1520	London, UK; 2011-2013[133]	Low	High	Low	Low	Low
p1557	Istanbul, Turkey; 2010-2011[135]	High	Low	Low	High	Low
p1612	Memphis, Nashville, Salt Lake City, US; 2010-2012[104]	Low	High	Low	Low	High
p1733	Ho Chi Minh, Vietnam; 2004-2008[137]	Low	Low	Low	High	Low

Table S6.7 Risk of bias for studies with data on proportion positives of human parainfluenza virus in hospitalised ALRI

Study ID	Location; period	Study design	Patient groups excluded	Case definition	Sampling strategy	Test
p52	Perth, Australia; 2000-2005[163]	High	Low	Low	High	Low
p101	Yaounde, Cameroon; 2011-2013[128]	Low	Low	Low	Low	Low
p216	Beijing and Shandong, China; 2012-2015[143]	Low	Low	Low	High	Low
p221	Beijing, China; 2007-2010[144]	Low	Low	Low	High	Low
p241	Beijing, China; 2011-2012[145]	High	Low	Low	High	Low
p264	Changsha, China; 2007-2008[146]	Low	Low	Low	High	Low
p311	Hebei, China; 2007-2008[147]	Low	Low	Low	Low	High
p367	Lanzhou,China; 2006-2009[149]	High	Low	Low	Low	Low
p376	multisites,China; 2009-2013[150]	Low	Low	Low	Low	Low
p386	Shanghai, China; 2003-2006[151]	High	Low	Low	High	High
p387	Shanghai,China; 2013-2015[152]	Low	Low	Low	Low	High
p396	Shantou, Shenzhen and Jieyang, China; 2007-2011[154]	Low	Low	Low	High	Low
p399	Shantou, Shenzhen and Jieyang, China; 2006-2008[155]	Low	Low	Low	High	Low
p400	nine cities, North China; 2015[156]	Low	Low	Low	High	Low
p404	Shenzhen, China; 2012-2015[157]	Low	Low	Low	High	High
p421	Tianjin, China; 2015-2016[158]	Low	Low	Low	High	Low
p466	multisites, Egypt; 2007-2014[90]	Low	Low	Low	Low	Low
p523	Kiel, Germany; 1995-1999[164]	Low	Low	Low	High	Low
p540	Athens, Greece; (pre 2003) [165]	Low	High	Low	Low	Low
p612	Odisha, India; 2012-2014[166]	Low	Low	Low	High	Low
p671	Naples, Italy; 2016-2017[169]	High	Low	Low	High	Low
p738	Kilifi, Kenya; 2007-2010[124]	Low	Low	Low	High	Low
p829	Multi-country; 2008-2009[175]	High	Low	Low	High	Low
p953	Lørenskog, Norway; 2012-2014[176]	Low	High	Low	Low	Low
p1018	Riyadh, Saudi Arabia; 2005-2010[129]	High	Low	Low	High	High
p1071	multisites, South Africa; 2009-2012[83]	Low	Low	Low	Low	Low
p1488	Seongnam, South Korea; 2006-2016[130]	High	Low	Low	High	Low
p1489	Seoul, South Korea; 1996-1998[131]	Low	Low	Low	High	High
p1492	Seoul, South Korea; 2006-2007[132]	Low	Low	Low	High	Low
p1520	multisies, Spain; 2011-2013[133]	Low	High	Low	Low	Low
p1520	London, UK; 2011-2013[133]	Low	High	Low	Low	Low
p1557	Istanbul, Turkey; 2010-2011[135]	High	Low	Low	High	Low
p1612	Memphis, Nashville, Salt Lake City, US; 2010-2012[104]	Low	High	Low	Low	High
p1733	Ho Chi Minh, Vietnam; 2004-2008[137]	Low	Low	Low	High	Low
p1870	Cairo, Egypt; 2010-2014[139]	Low	Low	Low	High	Low
p1873	Lanzhou, China; 2004-2005[140]	Low	Low	Low	Low	High
p1901	Chengdu, China; 2007[141]	Low	Low	Low	Low	High
p1931	Changsha, China; 2015[142]	Low	Low	Low	Low	High

Checklist of information that should be included in new reports of global health estimates

Item #	Checklist item	Reported on page #
Objectives and funding		
1	Define the indicator(s), populations (including age, sex, and geographic entities), and time period(s) for which estimates were made.	Summary
2	List the funding sources for the work.	Summary
Data Inputs		
For all data inputs from multiple sources that are synthesized as part of the study:		
3	Describe how the data were identified and how the data were accessed.	Page4
4	Specify the inclusion and exclusion criteria. Identify all ad-hoc exclusions.	Page5
5	Provide information on all included data sources and their main characteristics. For each data source used, report reference information or contact name/institution, population represented, data collection method, year(s) of data collection, sex and age range, diagnostic criteria or measurement method, and sample size, as relevant.	Appendix pp18-41
6	Identify and describe any categories of input data that have potentially important biases (e.g., based on characteristics listed in item 5).	Appendix pp42-55
For data inputs that contribute to the analysis but were not synthesized as part of the study:		
7	Describe and give sources for any other data inputs.	Appendix pp18-41
For all data inputs:		
8	Provide all data inputs in a file format from which data can be efficiently extracted (e.g., a spreadsheet rather than a PDF), including all relevant meta-data listed in item 5. For any data inputs that cannot be shared because of ethical or legal reasons, such as third-party ownership, provide a contact name or the name of the institution that retains the right to the data.	Data have been presented in the supplementary material. Data will be made available on Edinburgh Datashare (https://datashare.is.ed.ac.uk/) later.
Data analysis		
9	Provide a conceptual overview of the data analysis method. A diagram may be helpful.	Figure 1
10	Provide a detailed description of all steps of the analysis, including mathematical formulae. This description should cover, as relevant, data cleaning, data pre-processing, data adjustments and weighting of data sources, and mathematical or statistical model(s).	Page 6-7; Appendix pp5-7
11	Describe how candidate models were evaluated and how the final model(s) were selected.	Page 7
12	Provide the results of an evaluation of model performance, if done, as well as the results of any relevant sensitivity analysis.	Appendix pp8-14
13	Describe methods for calculating uncertainty of the estimates. State which sources of uncertainty were, and were not, accounted for in the uncertainty analysis.	Page 6.
14	State how analytic or statistical source code used to generate estimates can be accessed.	Major codes used in this study will be made available upon request.
Results and Discussion		
15	Provide published estimates in a file format from which data can be efficiently extracted.	Estimates can be easily extracted in main table and supplementary table. Main tables will also be provided on Edinburgh Datashare (https://datashare.is.ed.ac.uk/) later.
16	Report a quantitative measure of the uncertainty of the estimates (e.g. uncertainty intervals).	Uncertainty is reported for burden estimates in main tables, supplementary tables and Results.
17	Interpret results in light of existing evidence. If updating a previous set of estimates, describe the reasons for changes in estimates.	Page 13.
18	Discuss limitations of the estimates. Include a discussion of any modelling assumptions or data limitations that affect interpretation of the estimates.	Page 11-13

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